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Deep learning-based classification, detection, and segmentation of tomato leaf diseases: A state-of-the-art review

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ABSTRACT

The early identification and treatment of tomato leaf diseases are crucial for optimizing plant productivity, efficiency and quality. Misdiagnosis by the farmers poses the risk of inadequate treatments, harming both tomato plants and agroecosystems. Precision of disease diagnosis is essential, necessitating a swift and accurate response to misdiagnosis for early identification. Tropical regions are ideal for tomato plants, but there are inherent concerns, such as weather-related problems. Plant diseases largely cause financial losses in crop production. The slow detection periods of conventional approaches are insufficient for the timely detection of tomato diseases. Deep learning has emerged as a promising avenue for early disease identification. This study comprehensively analyzed techniques for classifying and detecting tomato leaf diseases and evaluating their strengths and weaknesses. The study delves into various diagnostic procedures, including image pre-processing, localization and segmentation. In conclusion, applying deep learning algorithms holds great promise for enhancing the accuracy and efficiency of tomato leaf disease diagnosis by offering faster and more effective results.

1. Introduction

Tomatoes are one of the most widely grown and economically significant crops worldwide. According to the Food and Agriculture Organization (FAO), global tomato production exceeds 180 million tons annually. This makes tomatoes a staple in both domestic and commercial agriculture. They contribute significantly to the economies of many countries. This is especially evident in regions where agriculture plays a major role in Gross Domestic Product (GDP) and employment. Moreover, tomato is among the most nutritious crops globally (Verma et al. (2020); Chen et al. (2020)). Tomatoes are not only nutritionally rich but also exhibit pharmacological effects. These effects contribute to health benefits by mitigating conditions such as hypertension, hepatitis, and gingival bleeding. Owing to its extensive applications, the demand for tomatoes is increasing (Wu et al. (2020); Agarwal et al. (2020)).

Despite their economic importance, tomato plants are highly susceptible to a range of diseases. Many of these diseases affect the leaves, including bacterial spot, early blight, and leaf mold. These diseases can lead to significant crop loss. They reduce both yield and quality. They impair essential processes like photosynthesis. This causes plants to wither and fruits to rot before ripening. The occurrence of such diseases is often influenced by climatic and weather fluctuations. Excessive or insufficient rainfall can affect tomato crops. Extreme temperatures also contribute to this vulnerability. These factors make tomato crops susceptible to viral, fungal, and bacterial infections (Joshi and Bhavsar (2020); Wang et al. (2020a); Mkonyi et al. (2020)). The global tomato industry faces challenges from these diseases, which not only reduce crop yield but also increase the costs associated with pest control and disease management. According to studies, tomato diseases can reduce production by as much as 20 % annually in severe cases. Early detection and accurate diagnosis are crucial to mitigating these losses and improving the sustainability of tomato farming. The world's population is expected to reach nine billion by 2050. Therefore, effective disease detection and management strategies are critical to increasing crop yield while minimizing pesticide use (Prabhakar et al. (2020); Sharma et al. (2020)).

Early and accurate identification of tomato leaf diseases is crucial. It helps farmers take timely action to prevent the spread of disease. This ensures higher yields and better quality crops. With the increasing demand for sustainable agriculture and the rise of precision farming,

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automating disease detection using AI and image processing is an innovative solution to a longstanding agricultural problem. Initially, researchers used manual techniques to extract features and classify diseases using simple methods, for example, back propagation networks, Support Vector Machine (SVM), Random Forest (RF), K-nearest neighbors (KNNs) and Naive Bayes (NB). Since 2015, the research has shifted to deep learning for feature extraction and classification. Deep learning automatically handles feature engineering through algorithmic optimization (Ngugi et al. (2020); Wang et al. (2020b); Karthik et al. (2020)). Current technologies predominantly rely on deep learning models, particularly convolutional neural networks. These models effectively diagnose and predict infections from crop images (Aravind et al. (2020); Xiong et al. (2020)). Integrating different segmentation techniques further enhances disease classification and detection (Lee et al. (2020); Nagaraju and Chawla (2020)).

This study reviews advancements in tomato leaf disease detection. It emphasizes the global significance of this issue. It highlights the critical need for cost-effective and accurate diagnostic methods. This need is driven by the growing demand for food security and the limitations of traditional manual detection approaches. Moreover, the review explores the transformative potential of deep learning models in plant disease management. It focuses specifically on their applicability to tomato plants. The study examines existing research to offer insights into scalable solutions that can be applied to diverse farming contexts, ranging from smallholder farms to large-scale agricultural operations.

The review focuses exclusively on identifying tomato leaf disease through deep learning techniques. It addresses a notable gap where most articles cover broader plant leaf diseases. A limited number of dedicated reviews specifically target tomato leaf diseases. This highlights the distinctiveness of this review. Unlike broader plant-centric studies, this dedicated study provides a more precise exploration of tomato leaf disease identification. It offers an in-depth analysis of the challenges and advancements using deep learning methods. Table 1 shows our article's comparative study of existing review articles. It evaluates whether each paper includes discussions on datasets, pre-processing techniques, explained algorithms, disease types, survey-wide analysis, challenges, and novel contributions. A "✓" indicates the topic is adequately addressed, while a "✗" signifies its absence.

The "Datasets Discussed" column identifies whether the authors of each paper have discussed the datasets used for tomato leaf disease diagnosis. The "Pre-processing Techniques Described" column determines if the authors have discussed commonly used preprocessing methods for classification, detection, or segmentation of tomato leaf diseases. In our survey, we reviewed papers on tomato leaf disease diagnosis using deep learning models. Therefore, we provided a separate section describing the preprocessing techniques found in these studies. The "Explained Models for Leaf Disease Diagnosis" column indicates whether the authors have covered widely used deep learning models for tomato leaf disease classification, detection, or segmentation. Based on our survey, we identified the most effective models and categorized them into three subsections: classification, detection, and segmentation. Additionally, we highlighted how authors have improved these models using various techniques. Notably, since this paper focuses on tomato leaf disease diagnosis, readers from various disciplines may seek information about different diseases affecting tomato leaves. Therefore, the "Tomato Leaf Diseases Explained" column specifies whether a paper includes a dedicated section discussing these diseases. The "Survey-wide Results Analysis" column reflects whether the authors have included an analysis of the model results from the collected studies. Furthermore, "Detailed Challenges & Future Research Directions" column examines whether the authors have discussed challenges in the reviewed studies and suggested future research directions. Lastly, the "Novel Contributions of Survey Paper" column outlines the unique contributions made by each survey.

Against this backdrop, this study provides novel insights and methodologies to further enhance the efficiency and accuracy of crop disease identification through innovative approaches in image processing and deep learning. The main contributions of this study are as follows:

- This review categorizes and examines a wide range of tomato leaf diseases, detailing their causes, symptoms, and effects. It provides an extensive overview of the different types of diseases, enhancing understanding of how various conditions impact tomato plants and their diagnosis.
- This review offers a thorough examination of various datasets used for tomato leaf disease detection, including their characteristics, quality,

Table 1
Comparative analysis of existing review articles with our article.

Ref.	Datasets Discussed	Pre-processing Techniques Described	Explained Models for Leaf Disease Diagnosis	Tomato Leaf Diseases Explained	Survey-wide Results Analysis	Detailed Challenges & Future Research Directions	Novel Contributions of Survey Paper
Saranya et al. (2021)	✓	✓	✓	✗	✓	✗	This review presents a comprehensive analysis of 38 research works applying deep learning techniques in various aspects of tomato plant research, emphasizing the influence of dataset quality on the performance of deep learning models.
Thangaraj et al. (2022)	✓	✗	✓	✗	✓	✓	This comprehensive review looks at how smart computer systems, especially deep learning, can help spot diseases in tomato leaves more accurately. It talks about the difficulties, solutions and useful ideas to make better predictions.
Shrivastav et al. (2023)	✓	✗	✗	✗	✓	✓	This study introduces a novel exploration into clustering-based segmentation for tomato plant disease detection, specifically incorporating swarm optimization techniques which offers promising avenues for enhancing system performance and accuracy.
Basha et al. (2023)	✗	✗	✓	Disease names mentioned only	✓	✓	This review explores how deep learning and machine learning can be used to identify diseases in tomato leaves, providing valuable insights for early detection and control in agriculture.
This study	✓	✓	✓	✓	✓	✓	This study contributes by providing a comprehensive exploration of tomato leaf diseases, including dedicated analysis of datasets, preprocessing steps, classification, segmentation and detection techniques, along with identifying best-performing models. It concludes with insights into challenges and future directions for improving tomato leaf disease identification.

and relevance. It highlights the strengths and limitations of these datasets, providing valuable insights for researchers selecting and utilizing data for deep learning models.

- Through extensive research, this study significantly contributes by exploring diverse preprocessing, advancing the precision and efficacy in identifying tomato leaf diseases. This research enhances the understanding and application of crucial techniques for accurate disease diagnosis.
- This work has made a significant contribution by reviewing a large number of recent journal articles. It also has offered a comprehensive examination of well-known deep learning methods for diagnosing tomato leaf disease. This facilitates the identification of optimal disease categorization and detection tactics by highlighting the most successful models and methodologies.

The remainder of this article is organized as follows: Section 2 offers a concise overview of the survey methodology. In Section 3, various types of tomato leaf diseases, their causes and their subsequent effects on the economy are explored. Section 6 describes the basic deep learning techniques employed to identify tomato leaf diseases. Section 4 addresses the datasets mentioned in the articles for classifying tomato leaf diseases from images using deep learning techniques. Popular preprocessing techniques has been described in section 5. The analysis results of articles collected from 2021 to 2023 are presented in Section 7. Section 8 presents the overall limitations of the reviewed studies and suggests possible future directions for overcoming them and section 9 concludes the paper.

2. Survey methodology

An elaborate two-step process was used to select relevant papers carefully. In the initial stage, duplicates and irrelevant articles were eliminated through thoroughly reviewing the titles and abstracts. Subsequently, the remaining papers underwent in-depth analysis. This allowed for a comprehensive extraction and summarization of relevant studies. The survey was confined to high-quality scholarly articles in databases such as ScienceDirect, SpringerLink, IEEE Xplore, Taylor and Francis, Wiley, Frontiers and Nature. Table 2 demonstrates the inclusion and exclusion criteria for selecting any paper to be part of the result and analysis section.

The paper selection process involved the formulation of a query comprising of three parts and specific keywords. The first part included keywords such as “Tomato Leaf Disease” and “Plant Leaf Disease,” connected by the “OR” operator. This ensured that the papers related to either term were considered. The second part incorporated the keywords “Classification,” “Detection,” and “Segmentation,” which focused on specific tasks within the leaf disease analysis. The connection between the second and third parts involved the “WITH” operator. It links the chosen task from the second part with the keywords from the third part. The third part encompassed terms, for example, “Deep Learning,” “CNN,” “Computer Vision,” and “Image Processing”. This finding highlights the technological approaches used in the selected studies. Through this structured query, the selection of papers aimed to cover a broad spectrum of research related to tomato leaf disease analysis. Examples of search queries included “Tomato Leaf Disease Classification WITH Deep Learning” or “Plant Leaf Disease Detection WITH Deep Learning”. Thus, it helps focus on specific topics of interest in computer vision. While searching for research on “Plant leaf disease classification/detection/segmentation with deep learning,” we identified several papers that included studies on various plants, including tomatoes. From these studies, we extracted and evaluated the performance results of deep learning models specifically on tomato leaves for the results and analysis section.

Table 2
Inclusion and exclusion criteria for selecting any paper.

	Inclusion Criteria	Exclusion Criteria
Types of study	Original and review articles	Thesis, white papers, communication letters, reports and editorials
Language	Research articles written in English	Non-English articles
Baseline	Deep learning	–
Publication year	Articles published from 2019 to 2023 (Result analysis part)	–
Source	Articles published in academic journals and conference papers	Book chapters

In the initial phase of this research, 480 papers were chosen for review. Fig. 1 provides an overview of the diverse keywords employed for article selection across various databases.

Following a comprehensive assessment, 112 papers were selected for the study. These papers explore the economic impact of tomato leaf diseases and assess the effectiveness of deep learning techniques in their identification. Papers featuring plants other than tomatoes were not entirely excluded. Instead, relevant information regarding tomato leaf disease classification was extracted and analyzed for model performance in the section 7. Here Fig. 2 shows the prisma flow diagram of article selection procedure according to this study.

3. Tomato leaf diseases

Fig. 3 presents the categorization of tomato leaf diseases into four types: fungal, viral, bacterial, and insect-borne diseases. Most of the images are sourced from the PlantVillage dataset (Hughes et al. (2015)), while the remaining images were obtained from various web sources (Reeves (2022); Subedi et al. (2015); Khethari (2024); Wintermantel and Wisler (2006)). In agriculture, plant diseases emerge because of climate fluctuations and diverse weather patterns in different regions (Liu and Wang (2020); Basavaiah and Arlene Anthony (2020)). Variables such as excessive or inadequate rainfall and extreme temperature contribute to the susceptibility of crops to viral, fungal and bacterial infections. These diseases greatly degrade crop quality (Pattnaik et al. (2020); Hernández and López (2020); Noon et al. (2020)). Attaining optimal crop quality and quantity is a formidable task for farmers to overcome these challenges. Identification of plant diseases remains an ongoing challenge. It affects both crop production and the overall plant growth (Zhang et al. (2020)). Hence, it is essential to thoroughly investigate various tomato diseases, their origins, their economic impacts and preventive measures. Table 3 concisely describes 14 tomato leaf diseases, their causes of infection and the influence of temperature and humidity on the emergence of these diseases.

3.1. Importance of early detection

Early detection of tomato leaf diseases is critical for several reasons:

- **Preventing Disease Spread:** Identifying diseases early allows for quick intervention, which can prevent them from spreading to healthy plants. For instance, early identification of ToMV-infected plants enables farmers to remove and destroy the affected specimens. This prevents the virus from spreading through mechanical means or vectors (Shankar et al. (2014); Rezazadeh, 2024).
- **Reducing Economic Losses:** Early intervention can significantly reduce economic losses by preserving crop yield and quality. Diseases like Late Blight can cause catastrophic losses if not managed promptly. Early detection enables the application of appropriate fungicides and cultural practices to reduce damage (Tiwari et al. (2021)).
- **Enhancing Treatment Effectiveness:** The effectiveness of treatment measures is often contingent on the disease stage at which they are applied. Early-stage treatments are generally more effective. They also require less intensive resource input compared to interventions

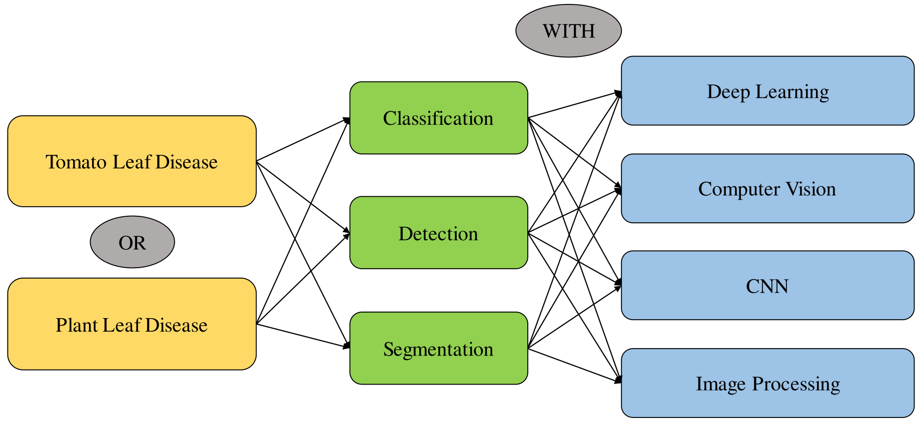

Fig. 1. Search mechanism or query for collecting relevant documents for review study.

at advanced disease stages. For example, applying fungicides at the onset of Early Blight symptoms can effectively control the disease progression (Cucak (2020)).

- **Promoting Sustainable Agriculture:** Timely detection of diseases contributes to sustainable agricultural practices. Proper management helps reduce the need for excessive chemical inputs. This minimizes the environmental impact of farming. It also promotes ecological balance.

3.2. Can we eliminate the disease once detected?

The ability to eliminate a tomato leaf disease once it has been detected depends on several factors, including the disease type, the stage of detection, and the available management strategies.

3.2.1. Disease type and detection stage

- **Viral Diseases:** Viral diseases, for example, ToMV and YLCV are challenging to eliminate once a plant is infected. Viruses integrate into the plant cells. They spread rapidly and make control difficult. If detected at an early stage, it is not possible to cure the infected plant. However, the spread of the virus can be controlled. This can be done by removing and destroying the infected plants immediately. This prevents further spread to nearby healthy plants.

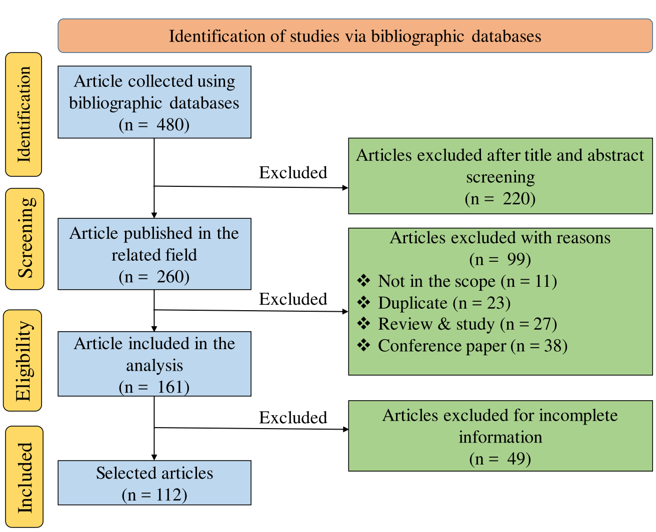

Fig. 2. Prisma flow diagram of the article selection procedure.

- **Fungal Diseases** Fungal diseases, for example, EB and LB can sometimes be managed effectively if detected early. Fungicides can halt or slow the progression of the disease. This is especially effective in the initial stages. However, if the disease is detected late, it may not be possible to fully eliminate the pathogen from the affected plants. The focus then shifts to containment. Efforts are made to minimize crop loss.
- **Bacterial Diseases:** Bacterial infections are difficult to eradicate from infected plants. Like viral diseases, early detection primarily helps in preventing the spread. It is not focused on curing the infected plants.

3.3. Addressing the challenge of verifying virus presence using visual features

Detecting plant viruses solely based on visual features presents significant challenges due to the following reasons:

3.3.1. Challenges

- **Symptom Overlap and Variability Symptom Similarity:** Many plant diseases and physiological stress conditions can produce similar visual symptoms. These symptoms include mottling, chlorosis, or leaf curling. Such similarities make it difficult to differentiate between viral infections and other issues. These issues may include nutrient deficiencies or fungal infections. This can lead to false positives or negatives when using visual detection methods alone.
- **Environmental Factors:** Variations in environmental conditions, such as light, temperature, and humidity, can influence symptom expression. This leads to variability in how symptoms appear. The symptoms may differ even within the same plant or among plants with the same infection.
- **Latency and Subtle Symptoms Latent Infections:** Some viruses, including ToMV, can be present in the plant. They may not cause immediate or obvious visual symptoms. The virus can spread unnoticed in such cases. This makes visual detection unreliable in the early stages of infection.
- **Subtle Symptoms:** Early-stage symptoms may be too subtle for human observers to detect accurately. Some image-based detection systems may also fail to identify these symptoms. This further complicates visual diagnosis.

3.3.2. Solution

- **Molecular Diagnostics:** Polymerase Chain Reaction (PCR) and Reverse Transcription PCR (RT-PCR) are highly sensitive techniques. They are also specific for detecting viral genetic material in plant tissues. These methods can confirm the presence of a virus. This confirmation is often based on visual symptoms identified by deep learning models (Hadidi et al. (2017)).

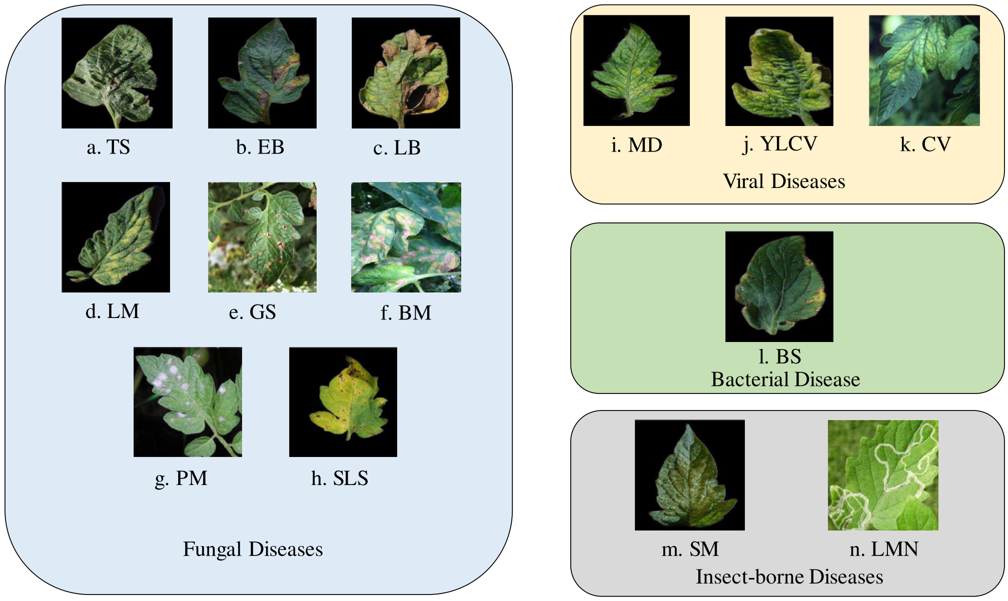

Fig. 3. Collection of tomato leaf disease images.

- **Serological Tests:** Enzyme-linked immunosorbent assay (ELISA) is another widely used method that can detect viral proteins in plant samples. It provides a reliable means to verify viral infections detected visually (Lister (1978)).
- **Spectral Imaging:** Hyperspectral imaging techniques can detect changes in leaf biochemistry and structure. Multispectral imaging techniques can also identify these changes. These changes are not visible to the naked eye or conventional cameras. These techniques enhance the accuracy of visual detection models by providing additional data layers (Mishra et al. (2017)).
- **Combination Approaches:** Deep learning models can be combined with molecular, serological, and spectral diagnostic methods. This integration enhances the accuracy of virus detection. It also significantly improves the reliability of the diagnostic process. This approach ensures that the visual features identified by AI are backed by biochemical evidence.

3.4. Economic viability of early detection and intervention

Deep learning-based early detection systems provide significant economic benefits. These benefits are outlined below:

- **Cost-Benefit Analysis:** The initial setup and training of deep learning models may involve costs. However, the long-term savings from reduced crop losses are significant. Optimized use of inputs often outweighs these expenses. Studies have shown that early detection systems can lead to a significant reduction in management costs for certain diseases (Dolatabadian et al. (2025)).
- **Increased Yield and Quality:** Timely interventions preserve both the quantity and quality of tomato yields. This leads to higher market value and profitability for farmers (Domingues et al. (2022)).
- **Reduced Labor and Resource:** Automated detection systems reduce the need for manual scouting. They enable precise application of treatments. This results in labor and resource efficiencies (Saini et al. (2025)).
- **Long-Term Sustainability:** Early detection contributes to sustainable farming practices. It helps minimize environmental impacts. It also promotes the resilience of agricultural systems. This has positive economic implications over time (Adewusi et al. (2024)).

3.5. Management strategies post-detection

- **Preventative Measures:** Early detection allows the implementation of preventative measures. Affected plants can be isolated to stop the spread of the disease. Protective fungicides or bactericides can be applied for better control. Improved cultural practices, such as better irrigation and crop rotation, can also help prevent the disease from spreading.
- **Resistant Varieties:** In the long term, planting disease-resistant varieties is one of the most effective strategies for eliminating the recurrence of certain diseases. This approach doesn't cure an existing infection but prevents future outbreaks.
- **Biological Control:** In some cases, biological control agents can be introduced to manage and reduce pathogen levels. While they may not eliminate the disease entirely, they help maintain it at manageable levels.

4. Datasets

The main section of this research explores datasets for identifying tomato leaf diseases. It highlights the use of deep learning techniques. These datasets provide a diverse and extensive collection of images. They enable the development and evaluation of robust models for accurate disease detection. Fig. 4 represents the datasets utilized for tomato disease identification with deep learning methods. Table 5 represents a brief details about the datasets used for tomato leaf disease identification.

Table 4 represents the tasks performed using various datasets, specifically classification, detection, and segmentation. A "✓" indicates that a dataset has been utilized for a particular task, while a "**x**" signifies that it has not. For instance, TLDD has been applied for detection, so a "✓" is placed in the detection column. Whereas "**x**" is assigned to both classification and segmentation, as it has not been used for these tasks. Some datasets have been employed for multiple tasks. For example, PlantVillage has been used for classification, detection, and segmentation. Although it was originally designed for classification, some researchers adapted it for other tasks. They achieved this by applying their own masking techniques for segmentation and creating bounding boxes for detection.

Table 3
Tomato leaf diseases and details.

Disease Name	Pathogen Type	Pathogen	Details
Tomato Target Spot (TS)	Fungus	*Corynespora cassiicola*	• First reported in 1972 in Florida (MacKenzie et al. (2019)) • Grows well during high humidity and temperatures between 83 °F and 90 °F • 16–44 h is required for spore germination • Manifests as small lesions (>1mm) with a yellow margin on the leaves, growing to 10 mm in the advanced stages • Infects all above-ground parts of tomato plants (Abdulridha et al. (2020))
Tomato Early Blight (EB)	Fungus	*Alternaria solani*	• Grows well during high humidity and temperatures between 24° to 29 °C • Manifests as small, dark necrotic lesions on older leaves • Severe infection can lead to premature defoliation • Particularly affects young plants (Chaerani and Voorrips (2006))
Tomato Late Blight (LB)	Fungus	*Phytophthora infestans*	• Exhibits a lighter ring around irregular lesions on younger leaves • White cottony growth on the undersides due to sporangia formation under high humidity (Soylu et al. (2006)) • As the disease progresses, lesions enlarge, leaves discolor, shrivel, and die • Affects particularly in humid regions with temperatures between 4 °C and 29 °C
Tomato Leaf Mold (LM)	Fungus	*Cladosporium fulvum*	• Thrives under warm and humid conditions • Primarily affects leaves but can extend to stems, fruits, and other tissues • Causes chlorotic spots, mold, curling, and withering • Targets young leaves, causing necrosis and reduced fruit quality • Causes significant yield losses (Zhao et al. (2022a))
Tomato Gray Leaf Spot (GS)	Fungus	*Stemphylium solani*, *S. lycopersici*, *S. botryosum*	• Causes small dark brown specks that develop into brown spots with a yellow halo • Spots eventually turn gray and may harbor gray molds • Leads to yellowing, drying, and leaf fall • Thrives in warm, humid climates with an optimum temperature of 24–27 °C • Disrupts photosynthesis
Tomato Black Mold (BM)	Fungus	*Pseudocercospora fuligena*	• Tiny, pale yellow lesions on both leaf surfaces that lack a clear border • Lesions generate white fungal growth, turning sooty black • Leaves curl prematurely upward • Prevalent in humid tropical climates (Mersha et al. (2014))
Tomato Powdery Mildew (PM)	Fungus	*Oidium neolycopersici*	• First observed in the UK in 1987 • Adaxial surface of tomato leaves develops powdery white lesions • May lead to infections of the calyx, petioles, and abaxial surfaces • Severe infections cause leaf chlorosis, early senescence, and reduced fruit size and quality (Jones et al. (2001))
Tomato Septoria Leaf Spot (SLS)	Fungus	*Septoria lycopersici*	• Exhibits circular to semi-circular spots with tan to gray centers and dark brown margins (Parker et al. (1997)) • Starting on the lower foliage, it progresses upward, affecting the stems and leaflets • Causes defoliation • Thrives in moist seasons, common in northeastern, Midwestern U.S., and southeast Canada
Tomato Mosaic Disease (MD)	Virus	*Tomato Mosaic Virus (ToMV)*	• First identified in the Netherlands in 1910 and in the USA in 1916 (Broadbent (1976)) • Affects young plants at 16–28 °C • Characteristics include light and dark green mottling, slight plant stunting, and occasional leaf distortion • Estimated 20 % loss in global tomato production attributed to tomato mosaic virus (Broadbent (1976))
Tomato Yellow Leaf Curl Disease (YLCV)	Virus	*Geminiviruses*	• Distinct leaflet curling • Halts plant growth and causes flower abortion • Thrives at 33 °C–39 °C and severely affects tomatoes • Can cause up to 100 % losses (Moriones and Navas-Castillo (2000))
Tomato Chlorosis Disease (CV)	Virus	*Tomato Chlorosis Virus (ToCV)*	• Discovered in Florida in the 1990s • Transmitted by whiteflies • Interveinal yellowing and leaf thickening, progressing from the base upward • Necrosis of older leaves affects plant vigor and fruit yield (Fiallo-Olivé and Navas-Castillo (2019))
Tomato Bacterial Spot (BS)	Bacteria	*Xanthomonas* species (*Xanthomonas euvesicatoria*, *X. vesicatoria*, *X. perforans* and *X. gardneri*)	• The pathogen thrives at temperatures ranging from 24 °C to 30 °C (Adhikari et al. (2020); Jones et al. (2004)) • Small dark lesions with a potential yellow halo characterize the disease • Spots concentrate on leaf edges and tips, growing 3–5 mm • Infected leaves may have a scorched appearance • Severe cases can lead to yellowing, defoliation, and plant decline • Potential yield loss of up to 66 %
Tomato Spider Mite (SM)	Insect	*Tetranychus urticae*	• Chlorotic lesions on tomato leaves, leading to yellowish-white and mottled foliage (Kant et al. (2004)) • Flourishes in low humidity and temperatures of 29 °C to 35 °C • In extreme cases, the entire plant can be killed by this disease
Tomato Leaf Miner (LMN)	Insect	*Tuta absoluta*	• Insects deposit eggs on the spongy layer of the leaf • Life span involves four stages: egg, larva, pupa, and adult • Larvae create tunnels in leaves, causing brown spots and reduced yields (Xu et al. (2007))

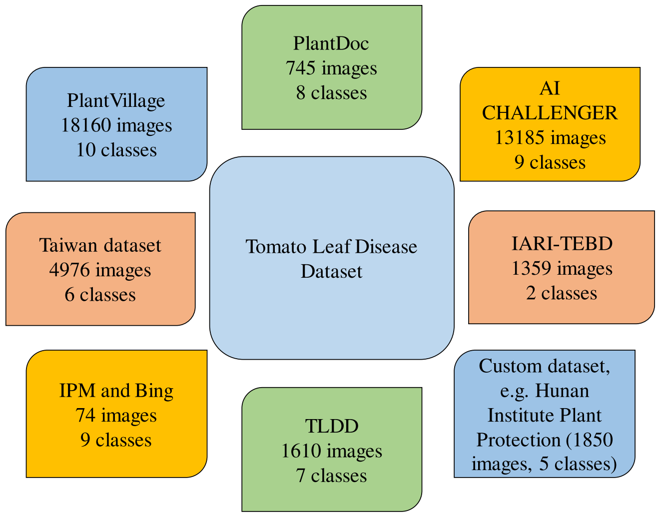

Fig. 4. Utilized datasets in tomato disease identification with deep learning methods.

4.1. Image acquisition methods

In the realm of deep learning-based image processing for agricultural applications, two prominent image acquisition methods are mobile phones and drones. Both methods present unique advantages and challenges:

- **Mobile Phones:** Mobile phones are accessible, user-friendly, and cost-effective. They allow for high-resolution, close-up images that are ideal for capturing detailed features of tomato leaves. This method is practical for both farmers and researchers. It enables frequent monitoring without requiring significant investment in specialized equipment.
- **Drones:** Drones offer the ability to capture images from a variety of angles and altitudes, covering large areas efficiently. They are particularly valuable for surveying fields and detecting patterns that may indicate the spread of disease. However, the use of drones for capturing tomato leaf images is still underexplored. Existing datasets are often limited, and the research conducted using drone-based images remains sparse. This presents a significant research gap in the current literature.

5. Preprocessing techniques

Preprocessing techniques are crucial in deep learning by preparing and refining data for effective model training. They help enhance the quality of input data, reduce noise and make it easier for the model to learn patterns and features. Proper preprocessing improves the performance of deep learning models. This leads to more accurate and reliable results. Table 6 shows detailed description of the preprocessing techniques.

Table 4
Comparison of datasets based on classification, detection, and segmentation capabilities.

Name of the Dataset	Classification	Detection	Segmentation
PlantVillage	✓	✓	✓
PlantDoc	✓	✓	✗
AI-CHALLENGER	✓	✓	✗
Taiwan dataset	✓	✗	✗
IARI-TEBD	✓	✗	✗
IPM and Bing	✗	✗	✓
Hunan Institute Plant Protection	✗	✓	✗
TLDD	✗	✓	✗

6. Tomato disease diagnosis with deep learning

Diagnosing tomato leaf disease employs a comprehensive approach that integrates classification, segmentation, and detection techniques. Classification identifies the disease type; segmentation precisely outlines the affected regions and detection pinpoints anomalies within the leaves. Here, Fig. 5 illustrates the tomato leaf disease model training workflow, encompassing classification, detection and segmentation.

6.1. Classification

The main goal of image classification in deep learning is to train a computer algorithm. Specifically, Convolutional Neural Networks (CNNs) are used to recognize and categorize objects or patterns in images. This is particularly crucial in tasks such as tomato disease classification, where the goal is to automatically diagnose and categorize diseases affecting tomato plants based on images. Image classification, which is the task of classifying images into one of several predetermined classes, is a key topic in computer vision. It is the foundation for additional computer vision tasks such as segmentation, detection and localization (Rawat and Wang (2017)). The workflow for tomato disease classification involves key steps, such as data collection, preprocessing, labeling, model selection, configuration, training, evaluation and fine-tuning. Data collection entails managing a diverse dataset of relevant images, whereas preprocessing involves resizing, standardizing, normalizing and augmentation. Data labeling assigns meaningful labels to input data for pattern learning. Pre-trained models, particularly those from ImageNet, are highly beneficial. They serve as efficient feature extractors, saving computational resources. Transfer learning enhances the model performance by applying knowledge from one classification task to another. The model configuration includes defining input layers, output classes, and hyperparameters. It is followed by training, evaluation, and potential fine-tuning for optimal performance. A successful classification requires a correct classification method and adequate training samples (Lu and Weng (2007)). The overall process aims to create accurate and efficient models for tomato disease identification in images.

6.1.1. AlexNet

AlexNet, introduced by Krizhevsky in 2012, marked a breakthrough in deep learning and image classification. This large, robust network has around 650,000 neurons and 60 million parameters (Hosny et al. (2020); Lv et al. (2020)). This pioneering CNN architecture won the ImageNet large scale visual recognition challenge by significantly reducing error rates (Strisciuglio et al. (2020)). With eight layers, including five convolutional and three fully connected layers, AlexNet utilizes Rectified Linear Units (ReLU) for activation and dropout for regularization. Its innovative use of parallelized Graphics Processing Unit (GPU) training accelerated model learning. AlexNet's success led to the widespread adoption of deep learning in computer vision. It influenced later neural network designs and contributed to the evolution of artificial intelligence. Fig. 6 shows the model architecture of AlexNet which is a leading CNN architecture in image classification tasks.

Researchers are conducting numerous ongoing experiments to modify and enhance the performance of the AlexNet model. Senbatu et al. (2022) adapted AlexNet for tomato leaf disease identification by expanding the network size to support a higher dropout rate. They also significantly increased the learning rate by 10-100× and raised the momentum to 0.95–0.99. They also applied maximum-norm regularization to control high weight values and improve model stability. On the other hand, Qiu et al. (2024) enhanced AlexNet by using HOG and LBP weighted fusion for feature extraction. This improvement boosted the efficiency and accuracy of image recognition. Thanammal Indu and Suja Priyadharsini (2022) optimized AlexNet using the Crossover based Wind-driven Optimized (CROWDO) algorithm. This approach combined a binary coding system with the WDO algorithm to

Table 5
Number of images per class in various datasets.

Name	Number of Images	Number of Images Per Class											
		EB	H	LB	YLCV	LM	ToMV	SLS	TS	TSSM	BS	PM	GS
PlantVillage	18,160	1000	1591	1909	5357	952	373	1771	1404	1676	2127	–	–
PlantDoc	745	85	65	120	75	90	50	150	–	–	110	–	–
AI CHALLENGER	13,185	792	1381	1569	4442	1126	–	1403	74	929	–	1469	–
Taiwan dataset	4976	–	848	784	–	536	–	–	–	–	880	1256	672
IARI-TEBD	1359	678	681	–	–	–	–	–	–	–	–	–	–
IPM and Bing	74	17	–	10	1	10	–	8	6	1	21	–	–
Hunan Institute of Plant Protection	1850	–	–	–	–	–	–	–	–	–	–	–	–
TLDD	1610	280	120	–	–	290	210	210	180	–	320	–	–

fine-tune AlexNet's hyperparameters. As a result, classification accuracy improved, and training time was reduced. Key hyperparameters, such as mini-batch size and dropout rate, are fine-tuned by the CROWDO algorithm. This fine-tuning improves performance without compromising accuracy.

6.1.2. XceptionNet

XceptionNet is a deep learning architecture that extends the ideas of the inception modules used in GoogLeNet (InceptionV1). It was first introduced by Chollet in 2017. The main innovation of XceptionNet is the replacement of regular convolutional layers with depth-wise separable convolutions. This change makes the model more efficient in terms of computational resources (Chollet (2017); Gülmez (2023)).

Fig. 7 shows the Xception architecture, a deep convolutional neural network well-known for its exceptional performance in image classification. The XceptionNet architecture balances model complexity and computational efficiency by using depthwise separable convolutions (Chen et al. (2021)). This design choice reduces the number of parameters while maintaining expressive power. It makes the model more computationally efficient than traditional architectures. In summary, XceptionNet extends the inception architecture. It emphasizes depthwise separable convolutions. This approach achieves a good trade-off between model complexity and efficiency.

6.1.3. EfficientNet

EfficientNet was introduced by Tan and Le in 2019. It uses a compound scaling method to improve performance. This method uniformly scales all network dimensions—depth, width, and resolution (Tan and Le (2021)). This approach ensures optimal model performance across various resource constraints (Huang and Liao (2023)). EfficientNet achieves state-of-the-art accuracy with fewer parameters, thus making it computationally efficient. Researchers are conducting numerous ongoing experiments to modify and enhance the performance of the EfficientNet model. Kotwal et al. (2023) optimized EfficientNet for tomato leaf disease classification through several methods. The author employed UNet to isolate disease-infected regions. The hunter-prey optimization (Hunt-PO) algorithm was used to fine-tune the model's weights. Essential features were extracted using scale-invariant feature transform (SIFT), Gray-Level Co-occurrence Matrix (GLCM), and Gabor filters. Finally, EfficientNet was optimized using a hybrid artificial driving algorithm (ADA). This algorithm combined artificial hummingbird optimization (AHBO) and driving training optimization (DTO). The approach improved classification accuracy and minimized errors. Nazir et al. (2023) optimized EfficientNet by incorporating a spatial-channel attention mechanism. This mechanism focused specifically on damaged areas. As a result, the recognition ability of the model improved significantly. The authors also added extra dense layers at the end of the model to enhance feature selection. These enhancements collectively boosted the model's performance.

Fig. 8 represents the EfficientNet, an optimal neural network for efficiency and scalability in performance. The architecture consists of a baseline network and a set of scaling coefficients to control the network's depth, width and resolution (Atila et al. (2021)). EfficientNet currently consists of models labeled from B0 to B7. Each of them have varied parameters ranging from 5.3 million to 66 million (Marques et al. (2020)). EfficientNet demonstrated superior performance in image classification tasks. It balanced accuracy and computational efficiency effectively. Its innovative design has made it a prominent choice for applications with limited computational resources, such as mobile and edge devices. This is achieved without compromising the model's predictive power.

6.2. Detection

To achieve a comprehensive understanding of an image, focus should be placed on categorizing various images. Additionally, accurately estimating the ideas and positions of items in each image is essential. This is called the object detection task. Tomato disease identification is the specialized task of recognizing and localizing different diseases within images of tomato plants. Unlike classification, detection involves accurately identifying and locating multiple diseases in an image. In addition, it provides detailed information about the precise location of each disease. The primary goal is to detect various diseases in a tomato plant image. Each disease is then assigned with a specific label for identification. The detection output includes bounding boxes outlining the diseases and their corresponding class labels (Zhao et al. (2019)). Common architectures for tomato disease detection involve Region-based CNN (R-CNN), Fast R-CNN, Faster R-CNN and Single Shot Multibox Detector (SSD) (Albahli and Nawaz (2022)). Managing an annotated dataset with labeled bounding boxes for diseases of interest is part of the data preparation process. A critical first step is selecting a suitable detection model based on specifications like speed and accuracy. To train the model, both the image data and the associated bounding box annotations must be incorporated into the annotated dataset. The compiled model utilizes loss functions, including classification loss for disease classification and localization loss for precise bounding box prediction. The deployment of the trained model involves inference on new, unseen tomato plant images and generating bounding boxes around the detected diseases and their associated class labels. Bounding boxes are improved by post-processing methods like non-maximum suppression, which removes duplicate or overlapping detections. Tomato disease detection provides spatial information about disease locations, allowing for accurate localization of various diseases within tomato plants. This approach is versatile in handling scenarios where multiple diseases of different classes coexist in the same image, making it a practical solution for the diagnosis of tomato disease.

6.2.1. CornerNet

CornerNet is a detection model specifically designed for object detection in computer vision tasks. CornerNet focuses on detecting object keypoints. Specifically, it identifies corners or keypoints that represent the extreme points of an object. This approach differs from traditional object detection models that rely on bounding boxes (Nguyen (2023)). CornerNet aims to improve localization accuracy and handle occlusions more effectively by detecting keypoints. The model utilizes a main feature extraction network to calculate keypoint maps, which

Table 6
Preprocessing techniques and details.

Preprocessing Technique	Description
Image resizing	Image resizing means converting input images to a specified size before feeding them into a neural network. This method guarantees consistency throughout the dataset and improves computing efficiency (Lin et al. (2014)).
Mean filter	Replaces each pixel in an image with the average value of its neighboring pixels. It helps reduce noise. It makes the image smooth by blurring small variations in intensity (Thanh et al. (2019)).
Gaussian filter	Gaussian filters are good for lowering high-frequency noise, for example, pixel intensity fluctuations at random (Ito and Xiong (2000)).
Median filter	The median filter is a non-linear filter with decent edge retention capabilities. It can eliminate impulse noise from the images. The noisy values are replaced by the median value of the neighborhood mask (Justusson (2006)).
Canny edge detection	Canny edge detection is used to detect edges and boundaries within an image. Its ability to produce accurate and well-defined edges makes it a popular choice in various image analysis tasks (Bao et al. (2005)).
Data normalization	Equal weight distribution for each variable is ensured through normalization. The maximum value of each attribute is adjusted to fall within the same [0–1] range as the highest value. Normalization lowers training error. It also enhances the categorization outcome (Finlayson et al. (1998)).
Contrast Limited Adaptive Histogram Equalization (CLAHE)	Technique which is used for the enhancement of the contrast of the images. CLAHE is also used to mitigate non-uniform illumination (Reza (2004)).
Color transformation	This technique involves manipulating the color characteristics of images. The primary purpose is to enhance certain features and to reduce the impact of noise. Familiar color spaces are RGB (Red, Green, Blue) (Saravanan (2010)), HSV (Hue, Saturation, Value) (Shuhua and Gaizhi (2010)), Grayscale conversion (Saravanan (2010)), etc. are common color transformation techniques.
Masking green pixels	Masking green pixels involves selectively isolating green pixels from an image based on a predefined criterion. The primary goal is to eliminate or reduce the influence of green elements in an image, allowing subsequent processing steps to focus on other features of interest (Jaware et al. (2012)).
Extraction of contour coordinates	The primary goal is to extract the contour information, which represents the spatial arrangement and boundaries of objects in the image. Contour information is valuable for recognizing and distinguishing different objects in the image (Wu (2007)).
Automatic labeling	Automatically assign accurate labels or categories to data instances without manual intervention (Novozámskỳ et al. (2020)).
Deep dream segmentation	Deep dream can enhance patterns and textures in an image. It can help visualize features that the neural network considers important (Sahu et al. (2023)).
Asymptotic non-local means	This technique leverages the redundancy present in natural images by comparing similar patches from different regions of the image. Instead of relying on local information alone, the algorithm considers non-local similarities to better preserve image details (Huang et al. (2020)).
Background removal	Background removal involves isolating the foreground objects in an image while eliminating or suppressing the background. Removing the background enhances the accuracy of object recognition algorithms (Lu and Guo (1999)).
Histogram stretching	Histogram stretching enhances the contrast of an image by expanding the range of intensity values. It improves the visibility of details in an image by utilizing the full dynamic range of pixel intensities (Oliver (1998)).
Dynamic range expansion	Dynamic range expansion is used to enhance the visibility of details in an image by expanding the range of intensity values. This method increases the contrast between different regions in the image. It makes it easier to distinguish various features (Masia et al. (2017)).
Histogram plot of intensity	Analyzing the histogram can provide insights into the overall contrast and distribution of pixel intensities in an image. Peaks and valleys in the histogram show the prevalence of certain intensity levels. This helps in understanding the image's characteristics (Sun et al. (2015)).
Auto thresholding	Auto thresholding is used to segment an image by automatically determining optimal intensity thresholds. The goal is to separate objects or regions of interest from the background based on differences in pixel intensities (Sahoo et al. (1988)).
Otsu's thresholding	Otsu's thresholding automatically determines an optimal threshold value to separate pixels in an image into two classes: foreground and background. Pixels with intensities below the threshold are assigned to the background, while pixels with intensities above the threshold are assigned to the foreground (Guo et al. (2018)).
Histogram equalization	The primary goal of histogram equalization is to make the pixel intensity distribution more uniform which leads to improved visibility of details in both dark and bright areas of the image (Pizer et al. (1987)).
Feature selection with the mRMR	Feature selection with the mRMR is used in machine learning and pattern recognition to choose the most informative and relevant features from a given set. It enhances model performance by reducing the dimensionality of the input data while retaining the most discriminative features (Toğaçar et al. (2020)).
Geometric image transformation	Geometric image transformation involves modifying the spatial arrangement of pixels in an image based on geometric principles. This transformation can be applied to achieve various objectives, such as correcting distortions, resizing, rotating or repositioning the image (Wolberg (1988)).

are then employed to predict heatmaps, embeddings, offset and class information (Nawaz et al. (2021); Albattah et al. (2023)). The model architecture was designed to predict the presence of an object. It also predicts the locations of the object's keypoints. This provides a more detailed and accurate representation of object boundaries.

Fig. 9 shows CornerNet, an algorithm that can tackle precise localization and occlusion challenges in object detection tasks. Unlike traditional object detection models that predict bounding boxes, CornerNet focuses on detecting object key points, specifically an object's corners or extreme points.

6.2.2. Faster RCNN

Faster R-CNN is a powerful object detection model that was introduced by Ren in 2015. It innovatively integrates Region Proposal Network (RPN) within the network. This enables the model to generate region proposals for potential objects. The model can then classify them in a unified framework (Ren et al. (2015)). This two-stage architecture significantly improves the speed and accuracy of object detection compared to its predecessors (Sun et al. (2018)). Faster R-CNN utilizes deep CNNs to extract features. This provides a strong solution for real-time object detection tasks. As a result, it has become a fundamental element in advancing contemporary computer vision applications.

Fig. 10 shows the Faster R-CNN, a renowned architecture seamlessly incorporating object detection through its attention-focused RPN module. Faster R-CNN is a deep learning-based method for automated localization and categorizing plant leaf abnormalities. The use of convolve filters assesses the structure of suspected samples and extracts reliable key points (Jiang et al. (2021)). In contrast to RCNN and Fast-RCNN, which use handcrafted approaches for feature computation, Faster R-CNN introduces RPN. This separate module effectively addresses the issues present in RCNN and Fast-RCNN by automating the feature extraction from input images. The innovative use of the RPN enhances the model's ability to learn discriminative image features. This makes it a robust solution for plant leaf abnormality detection (Nawaz et al. (2022)). Hou et al. enhanced Faster RCNN for apple tree leaf disease

Table 7
Analysis of research articles on tomato leaf diseases classification published from 2019 to 2023.

Ref.	Disease	Dataset	Pre-Processing Method	Model	Results	Limitations
Thangaraj et al. (2023)	BS, LB, EB, MV, YLCV, LM, SLS, TS, SM	PlantVillage	Image resizing, Dataset splitting, Weight initialization, Softmax layer modification	Modified-Xception based Multi- Level Feature Fusion (MX-MLF2)	Accuracy: 99.61 %, Precision: 99.55 %, Recall: 99.40 %	• Sensitive to hyperparameter choices, requiring careful tuning for optimal results. • Might lack interpretability, making it challenging to understand the decision-making process. • Often demands significant computational resources.
Bora et al. (2023)	BS, LB, EB, MV, YLCV, LM, SLS, TS, SM	Not mentioned	Color transformation, Masking the green pixels, Segmentation, Feature extraction, Feature selection	MNDLNN + RKMC	Accuracy: 99.84 %, Recall: 99.26 %,	• Assumption-Data Distribution Mismatch due to Multivariate Normality. • Sensitivity to Outliers. • The expressiveness may limit when it come for tasks that require hierarchical feature learning. • Requires substantial computational resources.
Kotwal et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, YLCV, MV	PlantVillage	Image resizing, Gaussian filtering	EfficientNet + U-Net	Accuracy: 100 % (BS), Precision: 100 % (BS), Recall: 100 % (BS)	• Unable to select the most discrimination features. • Computationally complex.
Sahu et al. (2023)	EB, BS, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage, IARI-TEBD	Deep Dream Segmentation, Data augmentation, Data annotation, Data normalization	DD-Effinet-B4-ADB	Accuracy: 98.89 % (BS), F1 score: 99.1 % (BS), Precision: 100 % (BS), Recall: 99 % (YLCV)	• Selecting the most effective augmentation technique is challenging • Model could underfit after a certain level of augmentation. • Resized image to 144*144 for hardware constraints. • Unable to analyze leaf bunch for disease forecast. • Struggles with multiple diseases in single-leaf.
Deshpande and Patidar (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	N/A	Generative Adversarial Network (GAN) + Deep Convolutional Neural Network (DCNN)	Accuracy: 98.81 %, Precision: 99 %, Recall: 99 %, F1-score: 99 %	• Absence of preprocessing does not guarantee optimal performance.
Sharma et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage, Kaggle	Image resizing, Image augmentation, Image normalization	Deeper Lightweight Multi-class Classification-Network (DLMC-Net)	Accuracy: 96.56 % (BS), Precision: 98 % (BS), Recall: 99 % (YLCV), F1-score: 98 % (LM)	• Imbalance in classes results in overfitting. • Can not to classify multiple diseases in single-leaf.
Chong et al. (2023)	EB, LB, BS, YLCV	PlantVillage	Image resizing	ResNet-50-SVM	Accuracy: 93.33 %	• Requires optimization of the number of training samples.
Li et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Image resizing	Lightweight Multiple Branches Residual Network (LMBRNet)	Accuracy: 99.93 %, Precision: 99.72 %, Recall: 99.66 %, F1-score: 99.69 %	• Enhanced visuals mitigate downsampling-induced information loss.
Zhang et al. (2023)	BS, EB, LM, SLS, YLCV	PlantVillage, Tomato base of the Hunan Institute of Plant Protection	Image resizing, Image normalization, Image denoising (asymptotic non-local means), Data augmentation	Multi-channel Automatic Orientation Recurrent Attention Network (MAORANet)	Accuracy: 97.66 %, Precision: 97 %, Recall: 96 %	• Limited recognition ability on small data sets. • Recognition of early disease areas that still need improvement. • Target occlusion by foreign objects can lead to low recognition accuracy.
Dhanalakshmi et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Image resizing, Image augmentation	Modified InceptionV3	Accuracy: 99.60 % (MV), Precision: 100 % (MV), Recall: 100 % (LM), F1-score: 98 % (MV)	• Inadequate datasets: transfer learning and data augmentation approaches. • No-ideal robustness: unsupervised deep learning model and multi-condition training. • Symptom variations: gathering the full range of variance and progressively enriching the dataset variety. • Image spectral variations.
Sunil et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Image resizing	ResNet-50 + Multilevel Feature Fusion Network (MFFN) + Adaptive Channel Spatial and Pixel Attention Mechanism (ACSPAM)	Accuracy: 99.50 % (CPAM-4 L), Precision: 100 % (CPAM-4 L), Recall: 100 % (CPAM-4 L), F1-score: 100 % (CPAM-4 L)	• Imbalanced dataset. • Similar disease features cause misclassification risk. • Real-time image optimization may limit applicability.
Anim-Ayeko et al. (2023)	EB, LB	PlantVillage	Image resizing, Image normalization, Data augmentation	ResNet-9	Accuracy: 99.25 %, Precision: 99.67 %, Recall: 99.33 %, F1-score: 99.33 %	• Ignores key environmental and soil factors.

(continued on next page)

Table 7 (*continued*)

Ref.	Disease	Dataset	Pre-Processing Method	Model	Results	Limitations
Islam et al. (2023)	EB, LB	PlantVillage	Image resizing, Rescaling, Data augmentation	ResNet-50	Accuracy: 98.99 %, Precision: 98.96 %, Recall: 99.05 %, F1-score: 98.98 %	• Doesn't compare, evaluate user satisfaction or impact on decision-making. • Dataset limitations may not cover variations.
Zhong et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Image resizing	Lightmixer	Accuracy: 99.3 %, Precision: 98.6 %, Recall: 98.3 %, F1-score: 98.4 %	• Model evaluation limited to tomato leaf images from PlantVillage dataset. • The proposed model uses a fixed network architecture and hyperparameter settings, which may not be optimal for different scenarios and tasks.
Sanida et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Image resizing	Visual Geometry Group 16-layer (VGG-16)	Accuracy: 99.23 %, Precision: 99.12 %, Recall: 99.29 %, F1-score: 99.20 %	• Relies on pre-trained weights from ImageNet, which may not capture the specific features of tomato leaf diseases. • Ignores environmental factors; may impact tomato plant diagnosis and treatment.
Astani et al. (2022)	BS, EB, LB, SLS, LM, SM, TS, MV, YLCV, PM, GS, BM	PlantVillage, Taiwan dataset	Image segmentation, Contrast enhancement, Shadow removal, Background removal, Noise removal, Image contrast enhancement,	Ensemble ensemble classification strategy + Multilayer Perceptron (MLP) + K-means clustering	Accuracy: 95.98 %	• Taiwan dataset challenges: noise, clutter, shadows, low quality, multiple leaves. • Enhancing image quality amid challenges is critical for classification accuracy. • Ensemble classifier risks overfitting without proper tuning.
Kaur et al. (2023)	YLCV, MV, TS, SM, SLS, LM, LB	PlantVillage, PlantDoc	Image Resizing, Image Augmentation	Modified InceptionResNet-V2	Accuracy: 98.92 %, Recall: 97.94 %	• Modifed InceptionResNet-V2 may have some drawbacks such as high computational complexity, overftting, or gradient problems.
Kurmi et al. (2022)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Histogram Plot of Intensity, Auto-Thresholding, Background Removal	Deep CNN + Global and local thresholding	Accuracy: 92.6 %	• The method may struggle with noisy or occluded leaf images. • The method may struggle to detect multiple diseases on a leaf.
Li et al. (2022)	LM, SLS, YLCV	PlantVillage	Image resizing, Cropping, Normalization	Fast WDBlock based GAN (FWDGAN) + B-ARNet	Accuracy: 98.75 %, Precision: 98.75 %, Recall: 98.75 %, F1-score: 98.75 %	• FWDGAN method is constrained by lower-resolution images, requiring higher resolution. • FWDGAN method may suffer from mode collapse, limiting image diversity. • FWDGAN method neglects generator parameter optimization, impacting efficiency and scalability. • FWDGAN uses ssim-test for image diversity, lacks clarity on task-specific diversity levels.
Mukherjee et al. (2022)	BS, EB, LB, LM, SLS, SM, TS	PlantVillage	Image augmentation, Background subtraction, Otsu thresholding	Modified Gray Wolf optimized MobileNetV2 + Otsu thresholding + Mask R-CNN	Accuracy: 98 %	• MobileNetV2 may lack depth for capturing complex leaf disease features. • Incomplete evaluation hinders understanding model strengths, weaknesses and misclassifications' sources.
Djimeli-Tsajio et al. (2022)	YLCV, MV, TS, SM, LM, SLS, LB, EB, BS	PlantVillage	Object recognition by RGB color, The Viola and Jones algorithm for the selection of tomato leaves in the scene	ResNet101 and ResNet152 + MLP	Accuracy: 98.3 %, F1-score: 97.91 %	• Dataset limitations may affect the representativeness of tomato varieties and conditions. • Robustness, scalability and practicality in real-world applications are not evaluated.
Ahmed et al. (2022)	YLCV, MV, TS, SM, LM, SLS, LB, EB, BS	PlantVillage	CLAHE, Runtime augmentation	MobileNetV2	Accuracy: 99.30 %	• PlantVillage dataset limitations include samples taken in controlled laboratory conditions. • Further experiments with varied backgrounds from field-captured tomato leaf images are needed. • Field images may introduce challenges like occlusion and background clutter, requiring advanced segmentation techniques. • The dataset used in the experiment contains only a single disease per sample. • Identifying multiple diseases within a single leaf remains a challenging task.

Table 7 (*continued*)

Ref.	Disease	Dataset	Pre-Processing Method	Model	Results	Limitations
Zhang et al. (2022)	LM, SM, YLCV	PlantVillage	Feature extraction	Mixed Attention and Markovian Discriminator (MMDGAN) + B-ARNet	Accuracy: 97.12 %, F1-score: 97.78 %	• MMDGAN-based augmentation struggles with complex backgrounds, yielding incomplete, fuzzy images. • Resource-intensive MMDGAN model may not suit practical applications. • MMDGAN model may struggle with higher-resolution images' fitting ability. • MMDGAN model may require more suitable evaluation metrics for assessment.
Islam et al. (2022)	BS, LB, SLS, TS, SM, YLCV	PlantVillage, Own dataset	Image resizing, Data augmentation, Data balancing	DeepD381v1	Accuracy: 98.86 % (Net 1), Precision: 99.86 % (Net 1), Recall: 98.86 % (Net 1), F1-score: 98.86 % (Net 1)	• Resource-intensive training due to numerous layers and parameters. • Performance may falter in real-time, demanding controlled conditions. • Models may overlook real-world variability and complexity factors.
Zhao et al. (2022b)	TS, SM, LM, SLS, LB, EB, BS, YLCV, MV	PlantVillage	Image augmentation	RIC-Net	Accuracy: 95.20 %	• The dataset's clarity and singularity may limit the model's generalization and practical use in complex, real-world agricultural scenarios. • The model lacks testing on field images with varied conditions.
Al-gaashani et al. (2022)	BS, LB, LM, SLS, YLCV	PlantVillage	Image resizing, Normalization, Data augmentation	MobileNetV2 and NASNetMobile	Accuracy: 97 %	• Images were obtained under controlled lab conditions, limiting generalization.
Paymode and Malode (2022)	TS, SM, LM, SLS, LB, EB, BS, YLCV, MV	PlantVillage, Own field dataset	Image filtering, Gray transformation, Image sharpening, Scaling, Data augmentation	VGG16	Accuracy: 95.71 %	• Study overlooks environmental factors, introducing potential noise and bias.
Thanammal Indu and Suja Priyadharsini (2022)	LM, EB, TS, YLCV, SM	PlantVillage	Image resizing, Data augmentation	CROWDO optimized AlexNet	Accuracy: 100 % (LM), Precision: 100 % (SM), Recall: 99 % (TS), F1-score: 100 % (LM)	• CROWDO algorithm relies on WDO, lacking global optimality.
Gonzalez-Huitron et al. (2021)	TS, SM, LM, SLS, LB, EB, BS, YLCV, MV	PlantVillage	Image resizing, Data augmentation	Xception, MobileNetV3, NasNetMobile, MobileNetV2	Accuracy: 100 % (Xception), Accuracy: 98 % (MobileNetV3), Accuracy: 84 % (NasNetMobile), Accuracy: 75 % (MobileNetV2)	• Applying CNN models on a Raspberry Pi 4 may compromise accuracy due to the limited processing power and memory of the device, requiring quantization and reduction in numerical accuracy. • Limited tomato diseases; may not address all challenges. • Class imbalance in dataset may impact classifier performance.
Cengil and Çınar (2022)	TS, SM, LM, SLS, LB, EB, BS, YLCV, MV, BM, GS, PM	Tomato diseases detection dataset in kaggle, Taiwan dataset	Feature selection with mRMR (Minimum Redundancy Maximum Relevance) algorithm	AlexNet, ResNet50, VGG16 + Linear Discriminant Analysis (LDA)	Accuracy: 100 % (TS), Precision: 100 % (YLCV), Recall: 100 % (LB), F1-score: 100 % (TS)	• The hybrid CNN model's robustness to real-field image challenges wasn't assessed.
Nandhini and Ashokkumar (2021)	BS, SLS, LB, MV	PlantVillage	Not mentioned	Improved Crossover-based Monarch Butterfly Optimization (ICRMBO) - InceptionV3, ICRMBO-VGG16	Accuracy: 99.94 % (ICRMBO-InceptionV3), Accuracy: 99.98 % (ICRMBO-VGG16)	• The ICRMBO-CNN framework may incur high computational cost and complexity due to the optimization of CNN parameters and the crossover operation. • The ICRMBO-CNN framework may struggle with noisy or low-resolution images.
Sachdeva et al. (2021)	TS, SM, LM, SLS, LB, EB, BS, YLCV, MV	PlantVillage	Geometric image transforms	DCNN + K-means clustering + Canny filter	Accuracy: 98.9 %	• The model overlooks causal agents and underlying mechanisms, focusing solely on visual leaf features.
Zhou et al. (2021)	EB, LB, LM, PM, SLS, SM, TS	AI CHALLENGER	Image resizing	Restructured Deep Residual Dense Network	Accuracy: 95 %	• Analysis of features learned by Restructured Deep Residual Dense Network model is not provided. • Reliance on the dataset may not capture all leaf disease variations.
Kaur and Bhatia (2019)	BS, LB, LM, SLS, TM, YLCV	PlantVillage	Data augmentation	ResNet	Accuracy: 98.8 %, Error Rate: 1.15 %, Recall: 98.%, Specificity: 99.8 %, Precision: 98.8 %, False Positive Rate: 0.1 %, F1-score: 98.8 %	• Insufficient image data • Poor generalization capability

Table 8
Analysis of research articles on tomato leaf diseases detection published from 2019 to 2023.

Ref.	Disease	Dataset	Pre-Processing Method	Model	Results	Limitations
Jing et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Mean filter, Gaussian Filter, Median Filter, Canny Edge detection, Extraction of contour coordinates, Automatic labeling	BC-YOLOv5 + Automatic Labeling Algorithm	Precision: 96.4 %	• Computationally intensive, requiring powerful hardware for real-time processing. • Performance heavily relies on the quality and quantity of the training data and it may struggle with diverse or inadequately labeled datasets. • Sensitivity to Hyperparameters.
Pal and Kumar (2023)	LB, BS, EB, MV, YV, LM, SLS, TS, YLCV, SM	PlantDoc, PlantVillage	Image scaling, Enhancement, Contrast adjustment	AgriDet + Multivariant-Grabcut Algorithm	Accuracy: 96.99 % (SLS), Recall: 91.37 % (BS)	• No integrated and labeled images from practical scenarios. • Unable to detect numerous diseases in a single image or multiple occurrences of the same disease in one image.
Roy et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Data annotation, Data augmentation using cycleGAN	Principal Component Analysis (PCA) + DeepNet + F-RCNN	Accuracy: 99.60 %, Precision: 98.55 %, Recall: 98.49 %	• Reducing the proposed 10-layer CNN architecture addresses computational complexity while maintaining accuracy.
Albahli and Nawaz (2022)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Feature selection	DenseNet-77 based CornerNet	Accuracy: 99.98 %, Precision: 99.62 %, Recall: 99.53 %, F1-score: 99.57 %	• The approach exhibits limited detection performance for images with significant angular variations.
Raja Kumar et al. (2023)	EB, PM, LM, YLCV, SM, TS, LB, SLS	AI CHALLENGER	Noise removal, Data annotation, Data augmentation, Data resize	MRCNN + Ant colony optimizer	Accuracy: 97.66 %, Precision: 97 %, Recall: 96 %	• Data augmentation's complete effectiveness in addressing data limitations requires investigation.
Nawaz et al. (2022)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Not mentioned	Faster-RCNN	Accuracy: 99.97 %, Precision: 99.48 %, Recall: 99.38 %, F1-score: 99.42 %	• Reliance on PlantVillage dataset may limit generalization to diverse conditions. • Lack of analysis on computational cost, memory usage and power.
Kaur et al. (2022)	TS, LM, SLS, BS, EB, MV	Tomato Leaf Disease Dataset (TLDD)	Noise reduction, Histogram stretching, Digital light subtraction, Dynamic range expansion for contrast enhancement and normalization	Modified Mask R-CNN	Accuracy: 98 %, F1-score: 91.2 %	• Certain lesions are misidentified, significantly impacting the plant's health. • The reliance on precise ground-truth data annotations can be limiting. • The identification of multiple stresses is not addressed.
Gajjar et al. (2021)	TS, SM, LM, SLS, LB, EB, BS, YLCV, MV	PlantVillage	Image resizing, Data augmentation, Image annotation	CNN + SSD	Accuracy: 96.88 %, Precision: 93.78 %, Recall: 94.91 %, F1-score: 95.34 %	• Environmental factors, such as, weather, soil and pests are not considered. • The study's reliance on Nvidia Jetson TX1 may limit compatibility. • Economic feasibility, scalability and usability in diverse scenarios are unexplored aspects.
Fuentes et al. (2021)	Canker, LM, GM, YLCV, PM	Own dataset	Data augmentation	Faster R-CNN	mAP: 93.37 %	• Imbalanced class samples may impact model learning and recognition performance. • Limited dataset variations may hinder model generalization to diverse scenarios. • Class selection based on data availability may introduce bias and noise.

detection. The author integrated Feature Pyramid Networks (FPN) and Inception Squeeze-and-Excitation ResNet (ISResNet). This combination improved feature extraction and detection accuracy (Hou et al. (2023)).

6.2.3. YOLO

The YOLO (You Only Look Once) algorithm is a real-time object detection system that revolutionized the way object detection is performed in computer vision. Unlike traditional methods, YOLO treats object detection as a single regression problem. This approach eliminates the need for multiple passes over an image. As a result, YOLO is significantly faster and more efficient. Fig. 11 represents the YOLO model architecture. The core idea behind YOLO is to divide the input image into a grid. It then predicts bounding boxes and class probabilities for objects within each grid cell simultaneously. Each grid cell is responsible for detecting objects whose center falls within the cell. It produces a set of bounding boxes with confidence scores. The algorithm then filters out boxes with low confidence using non-max suppression to ensure that only the most relevant boxes are kept. One of the key advantages of YOLO is its speed. This makes it ideal for real-time applications such as autonomous driving, surveillance, and robotics. It balances accuracy and speed effectively. Earlier versions, such as, YOLOv1 and YOLOv2 sacrificed some precision compared to slower methods

Table 9
Analysis of research articles on tomato leaf diseases segmentation published from 2019 to 2023.

Ref.	Disease	Dataset	Pre-Processing Method	Model	Results	Limitations
Mzoughi and Yahiaoui (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage, IPM, BING	Data annotation, Data augmentation	PspNet 101 + EfficientNet	Precision: 97 % (YLCV), Recall: 97 % (SLS), F1-score: 97 % (YLCV)	• Bias in the PlantVillage dataset, which may limit the generalizability of the results. • Limitations in accurately identifying subtle disease symptoms in segmentation. • Struggles with unseen species in generalization. • Study neglects disease severity, affecting accuracy based on symptoms.
Shoaib et al. (2022)	EB, LB, LM, SLS, SM, TS, YLCV, BS, MV	PlantVillage	Noise removal, Foreground segmentation, Refine the leaf area and extract the homogeneous region of interest	Inception Net + modified U-Net	Accuracy: 99.91 %, Precision: 99.31 %, Recall: 99.29 %, R1-score: 99.30 %	• Untested on embedded systems, limiting practical deployment and scalability potential. • Framework's effectiveness relies on accurate segmentation and classification models, potentially impacted by environmental factors.
Huang et al. (2023)	BS, EB, LB, LM, SLS, MV, YLCV	PlantVillage	Background remove, Data augmentation	Fully Convolutional–Switchable Normalization Dual Path Networks (FC-SNDPN)	Accuracy: 97.59 %, F1-score: 98.75 %	• Lacks diverse backgrounds for accurate tomato leaf disease identification.
Khan et al. (2023)	BS, EB, LB, LM, SLS, SM, TS, MV, YLCV	PlantVillage	Image resizing, *Z*-score normalization, Data augmentation	Modified UNet + HPOARA	Accuracy: 98.2 %, Precision 98.34 %, Recall 98.34 %, F1-score: 98.06 %	• Overfitting issue. • Inadequate trainig data.
Deng et al. (2023)	BS, EB, LB, LM	PlantVillage	Image resizing, Data augmentation	MCUNet	Accuracy: 91.32 %	• Complex background issue.

like Faster R-CNN. However, the introduction of YOLOv3 and YOLOv4 brought significant improvements in accuracy. The latest YOLOv5 versions further enhance accuracy while maintaining speed.

YOLOv1 (2016): Introduced single-shot object detection, processing images in one pass (Diwan et al. (2023)).

YOLOv2 (2016): Added anchor boxes and multi-scale detection for better accuracy and small object detection (Jiang et al. (2022)).

YOLOv3 (2018): Improved feature extraction with Darknet-53 and multi-scale predictions for complex datasets (Jiang et al. (2022)).

YOLOv4 (2020): Enhanced accuracy and speed are achieved with Cross-Stage Partial DarkNet53 (CSPDarknet53) and Spatial Pyramid Pooling (SPP). Additional improvements come from the Path Aggregation Network (PANet) and Spatial Attention Module (SAT) techniques (Jiang et al. (2022)). CSPDarknet53 enhances feature extraction and gradient flow. It reduces computational costs by using partial connections between stages. Spatial Pyramid Pooling improves detection performance by pooling features at multiple scales. This approach captures information from different spatial resolutions. PANet enhances feature fusion in the detection process. It improves object localization through bottom-up path augmentation. It also employs adaptive feature pooling for better accuracy. SAT boosts model performance by focusing on relevant spatial regions and suppressing less important features.

YOLOv5 (2020): Optimized for efficiency with flexible model sizes and improved training techniques (Jiang et al. (2022)).

YOLOv6 (2022): Further improved backbone and inference speed for practical applications (Hussain (2024)).

YOLOv7 (2022): Advanced optimizations for speed and accuracy with new object detection techniques (Hussain (2024)).

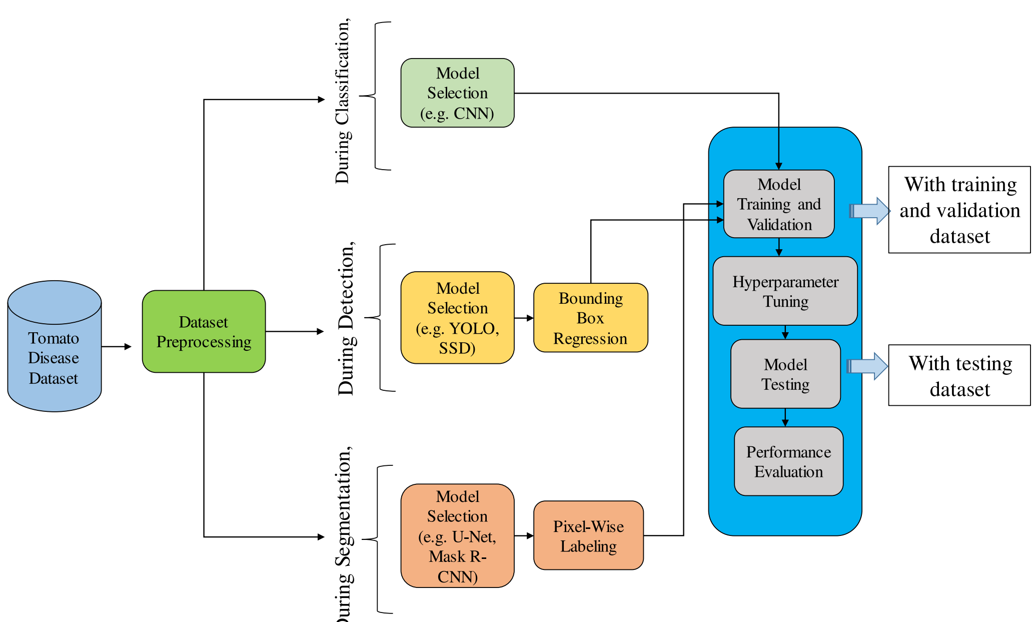

Fig. 5. Tomato leaf disease model training workflow, encompassing classification, detection and segmentation.

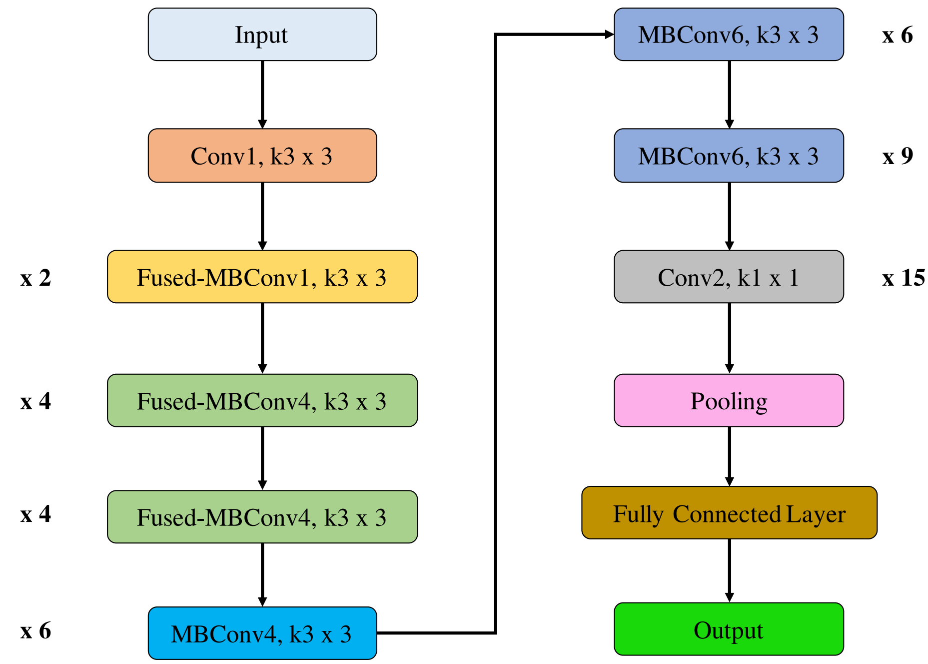

Fig. 6. AlexNet: pioneering CNN architecture.

YOLOv8 (2023): Signifies major evolution in the YOLO series, offering enhanced and versatile capabilities (Hussain (2024)).

YOLOv9 (2024): YOLOv9 features two key innovations. One of them is the Programmable Gradient Information (PGI) framework. The other one is the Generalized Efficient Layer Aggregation Network (GELAN) (Alif and Hussain (2024)). PGI enhances training efficiency and convergence by optimizing gradient flow and adaptively managing learning rates during model training. On the other hand, GELAN improves model performance. It also enhances computational efficiency. This is achieved by dynamically aggregating features across layers. This aggregation focuses on both spatial and channel information.

YOLOv10 (2024): This version advances real-time object detection. It is crucial for applications like agricultural monitoring and autonomous vehicle navigation (Alif and Hussain (2024)).

6.2.4. Multivariate normal deep learning neural network (MNDLNN)

In particular, the MNDLNN represents a state-of-the-art architecture at the forefront of tomato plant disease detection. The MNDLNN was created by carefully combining various deep learning approaches. Its complex design makes it useful for diagnosing illnesses. The input layer serves as the foundation. It manages the prepared image data. It also identifies the crucial elements that lay the groundwork for additional investigation. The hidden layers are the learning center of any

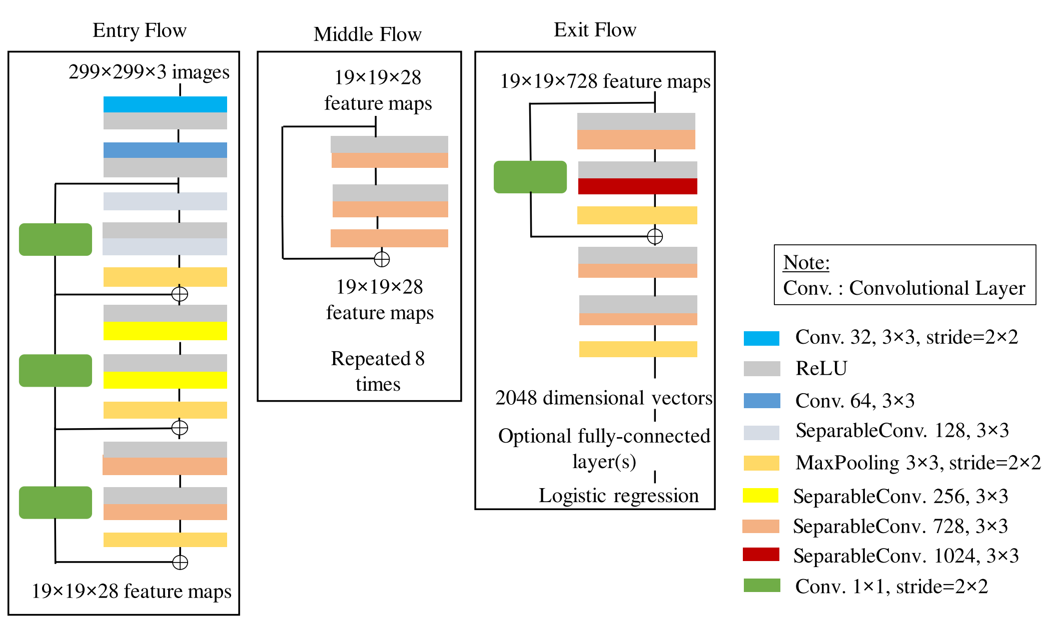

Fig. 7. The architecture of XceptionNet, a deep convolutional neural network known for its exceptional performance in image classification Chollet (2017).

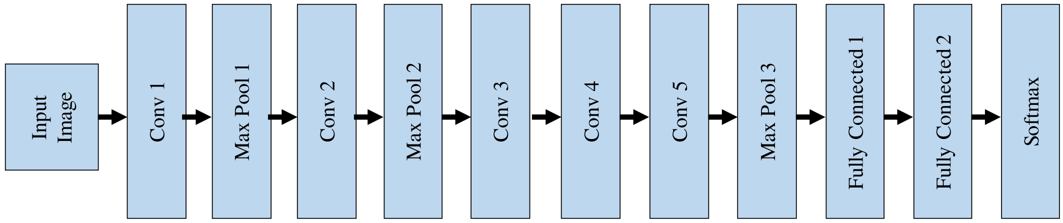

Fig. 8. EfficientNet: an efficient and scalable neural network architecture for optimal model performance.

neural network. They are enhanced with numerous neurons and functions. These enhancements enable the network to interpret complex patterns in the input data. What sets MNDLNN apart is its integration of advanced techniques, such as Rectilinear K-means Clustering (RKMC) and Random Motion Squirrel Search Optimization (RMSSO). These methodologies help the network work better by understanding the patterns in images with diseases. As the network progresses, the output layer takes center stage. It employs its classification process to deliver precise predictions regarding the nature of the diseases present.

The unique strengths of the MNDLNN extend beyond its technical architecture. MNDLNN is good at identifying and sorting diseases. It can be used not only just for tomatoes but also for other plants. Its multifaceted contributions span the realms of precision agriculture and global food security. This establishes the MNDLNN as a pivotal force in the ongoing battle against plant diseases.

6.3. Segmentation

In tomato disease segmentation, the process involves dividing the images into distinct segments based on predefined criteria. Unlike image classification, which labels the entire image, segmentation focuses on identifying specific items or regions. The goal is to understand the boundaries and structures of the various elements within an image. Segmentation techniques such as semantic and instance segmentation are used in deep learning for tomato disease classification. Semantic segmentation assigns a class label to each pixel. This process distinguishes the different disease categories in a tomato leaf image. Instance segmentation differentiates individual disease instances and provides a unique label for each instance (Minaee et al. (2021)). Integrating segmentation with tomato disease classification includes preprocessing, defining the model architecture and training. Sophisticated models such as Mask R-CNN and U-Net can be integrated by combining segmentation-specific layers with convolutional neural networks. This joint training enhances disease classification decisions. It provides improved localization and understanding of the disease composition and boundaries. Despite the added complexity, segmentation significantly boosts tomato disease classification performance. This improvement is especially evident in scenarios requiring accurate disease localization and object-level comprehension.

6.3.1. U-net

U-Net is a CNN architecture commonly used for semantic segmentation tasks in computer vision. U-Net is characterized by a U-shaped architecture. It includes a contracting path for context extraction and an expansive path for precise localization (Ibtehaz and Rahman (2020)). The network captures hierarchical features at different scales through the encoder and decoder pathways. This makes it effective for segmenting objects in the images (Ahmad et al. (2021)). U-Net has found widespread application in medical image analysis, particularly in tasks such as tumor segmentation in medical scans. Its design accurately defines object boundaries. This makes it a popular choice for various segmentation challenges.

Fig. 12 demonstrates the architecture of U-Net, which is a specialized deep-learning model for image segmentation featuring an encoder-decoder structure with skip connections for accurate object localization. U-Net is a CNN architecture commonly used for semantic segmentation tasks. Its distinctive U-shaped design consists of an encoder and a decoder. The encoder extracts hierarchical features through convolutional layers and reduces spatial dimensions. The decoder then reconstructs the original resolution by employing upsampling. Skip connections connect the equivalent layers between the encoder and decoder. This preserves fine-grained details during the upsampling process

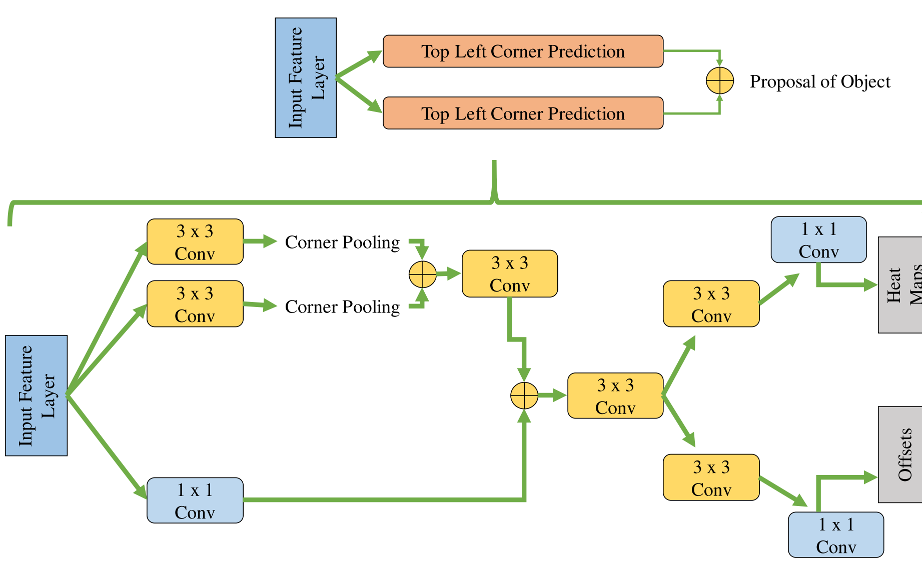

Fig. 9. CornerNet: an object detection model designed to address the challenges of precise localization and occlusion handling Nguyen (2023).

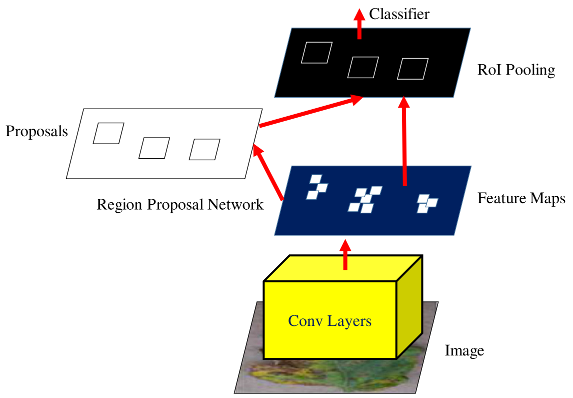

Fig. 10. Faster R-CNN unifies object detection with the RPN module serving as attention in the network Ren et al. (2015).

(Saeedizadeh et al. (2021)). This architecture is renowned for its effectiveness in accurately segmenting objects or regions of interest within images. This makes it a popular choice in various computer vision applications. Researchers are conducting numerous ongoing experiments to modify and enhance the performance of the U-Net model. Deng et al. (2023) proposed several enhancements to improve U-Net's performance for tomato leaf disease segmentation. The author introduced a Multi-scale Convolution Module (MCM) with varying convolution kernel sizes. This helps capture spatial information and address blurred edges in disease images. A Cross-layer Attention Fusion Mechanism (CAFM) was added to integrate features from different layers. This emphasizes disease areas while suppressing irrelevant details. Additionally, replacing MaxPool with SoftPool helped retain more disease features. Using the SeLU activation function ensured neuron stability by maintaining zero mean and unit variance in the sample distribution. El-Assiouti et al. (2023) introduced Lite-UNet, a lightweight architecture that enhances U-Net performance. By using MobileNetV2 as the encoder, Lite-UNet reduces parameter count and floating-point operations (FLOPs). Skip connections are incorporated to transfer information between layers. This improves segmentation mask accuracy.

6.3.2. Pyramid scene parsing network (PSPNet)

PSPNet is a deep learning model designed for semantic segmentation tasks. It particularly focuses on understanding scene-level information. PSPNet was first introduced in 2017. It improves accuracy by using a pyramid pooling module to capture multi-scale contextual information. PSPNet has a pyramid pooling module that aggregates information from multiple spatial scales. This allows the model to capture both global context and fine details. It enhances feature representation by pooling features at different scales. These pooled features are then concatenated to form a more comprehensive scene understanding. The model utilizes a fully connected CRF (Conditional Random Field) to refine the segmentation results and improve boundary precision.

PSPNet101 specifically refers to the PSPNet architecture that uses ResNet-101 as its backbone. ResNet101 is a deeper version of the ResNet architecture. It consists of 101 layers. Due to the deeper network, PSPNet101 can capture more complex features. This capability potentially leads to better performance in terms of accuracy. However, this comes at the cost of increased computational complexity and memory usage. PSPNet101 is used when higher accuracy is desired. It requires sufficient computational resources to handle the increased demands of a deeper network.

6.3.3. Leaf segmentation fuzzy CNN (LSFCNN)

Detecting a leaf within a complex background presents significant challenges, as multiple objects may be present at various locations. Factors such as lighting variations, shadows, and other environmental influences can result in non-uniform leaf colors. Additionally, the edges of tomato leaves vary significantly. This variation leads to different gradient changes from one leaf to another. The pixel color of the leaf may closely match the background, including stems or parts of other leaves. This makes it difficult even for humans to annotate the exact object. As a result, inconsistent annotations may occur. These annotation uncertainties can cause segmentation results to be unreliable. To address

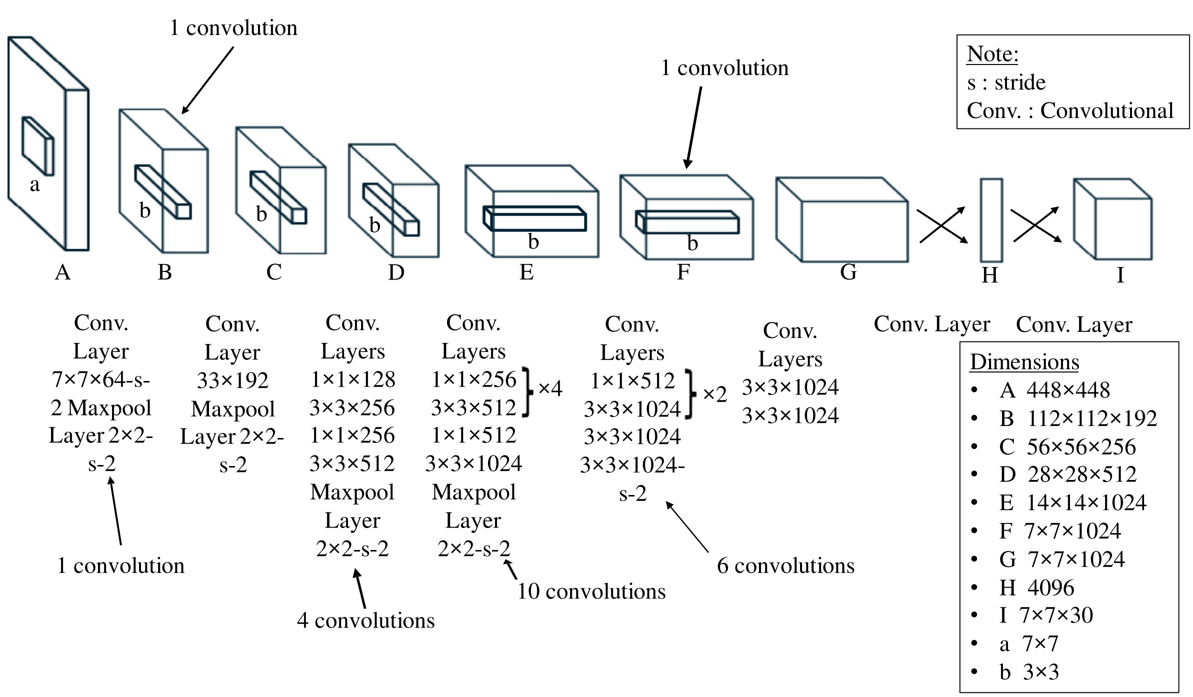

Fig. 11. YOLO model architecture Diwan et al. (2023).

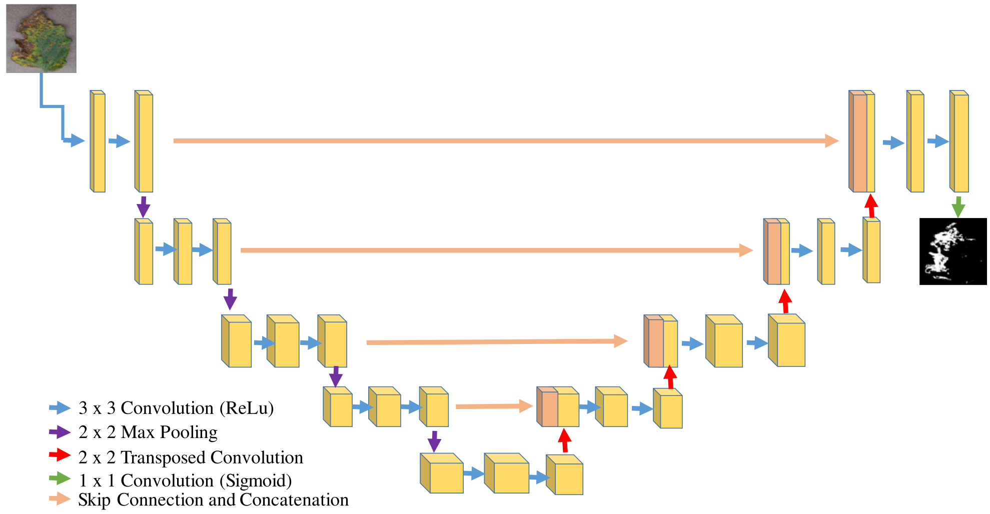

Fig. 12. The U-Net architecture integrates an encoder and a decoder pathway with skip connections linking corresponding layers.

this issue, LSFCNN-based segmentation integrated fuzzy learning with a deep neural network (Inception v2). Fuzzy models provide mathematical representations of ambiguity and imprecise data. They allow for the analysis of uncertain data across different levels. By incorporating fuzzy learning into various stages of the neural network, the model effectively handles uncertainties in the feature map. This results in an optimized feature map. The optimized map ensures accurate leaf segmentation. Fig. 13 represents the LSFCNN model diagram.

7. Results analysis

In this section, a comprehensive review of 48 papers that focused on the identification of tomato leaf diseases using deep learning techniques has been presented. Each study was scrutinized to determine the specific tomato leaf diseases that were addressed. The review includes insights into the employed models and their performance metrics such as accuracy, recall, precision, specificity mean Average Precision (mAP), etc. Additionally, we highlighted the datasets utilized by each study and identified whether they focused on detection or segmentation tasks. The summarized section offers insights into the models' performance, paving the way for future improvements. The subsequent analysis summarizes the best and worst-performing models based on the obtained results.

7.1. Overview: Result analysis of deep learning classification algorithms

This study aimed to provide a detailed analysis of the performance of these models in addressing various challenges related to tomato leaf disease identification. It encompasses a comprehensive exploration of 14 tomato leaf diseases. In comparison of these studies, the PlantVillage dataset was used 30 times. The Taiwan dataset was used twice. The IARI-TEBD, Hunan Institute of Plant Protection, PlantDoc, and AI Challenger datasets were each mentioned once. It demonstrated the use of diverse datasets to address various study concerns. Most of the studies used evaluation metrics such as accuracy, precision, recall, and F1-score. Gonzalez-Huitron et al. (2021) asserted that XceptionNet achieved perfect accuracy (100 %) in classifying ten classes. These included nine tomato leaf disease classes and one healthy class. They utilized the PlantVillage dataset with data augmentation. EfficientNet demonstrated 100 % accuracy for bacterial spots, as reported by Kotwal et al. (2023). Similarly, CROWDO-optimized AlexNet achieved 100 % accuracy for leaf mold disease, a finding supported by Thanammal Indu and Suja Priyadharsini (2022). Table 7 represents analysis of research articles on tomato leaf diseases classification published from 2019 to 2023.

7.2. Overview: result analysis of deep learning detection algorithms

In a comparison of tomato leaf disease detection studies, the PlantVillage dataset was most frequently used. It appeared six times. PlantDoc and AI Challenger datasets were each mentioned once, while one study utilized a customized dataset. Most of the studies used evaluation metrics such as accuracy, precision, recall, F1-score, and mAP. Tomato leaf disease detection studies employed various algorithms. These included BC-YOLOv5, AgriDet, F-RCNN, DenseNet-77 based CornerNet, Mask R-CNN (MRCNN), Modified Mask R-CNN, and SSD. These algorithms showcased diverse approaches in the research. According to Albahli and Nawaz (2022), a tomato leaf disease detection model called DenseNet-77 with CornerNet achieved the highest accuracy of approximately 99.98 %. Fuentes et al. (2021) utilized their own dataset for detecting tomato leaf diseases, including, LM, GM, YLCV, and PM. They applied data augmentation and employed Faster R-CNN. This approach achieved a mean Average Precision (mAP) of 93.37 %. Table 8 represents analysis of research articles on tomato leaf diseases detection published from 2019 to 2023.

7.3. Overview: result analysis of deep learning segmentation algorithms

In a comparison of tomato leaf disease segmentation studies, the PlantVillage dataset was most frequently used. It appeared five times. IPM and BING datasets were each mentioned once. Most of the studies used evaluation metrics such as accuracy, precision, recall, F1-score. Tomato leaf disease segmentation studies employed various algorithms, including UNET, modified UNET, FC-SNDPN, PSPNet, and MCUNet. These studies showcased diverse approaches in the research. A modified UNET, coupled with InceptionNet, achieved an impressive accuracy of 99.91 % in classifying 10 classes within the PlantVillage dataset. This result was reported by Shoaib et al. (2022). Table 9 represents analysis of research articles on tomato leaf diseases segmentation published from 2019 to 2023.

8. Challenges and future research directions

Fig. 14 represents the challenges and future research directions of tomato leaf disease identification. Tomato leaf disease classification using deep learning models reveals various challenges in this section.

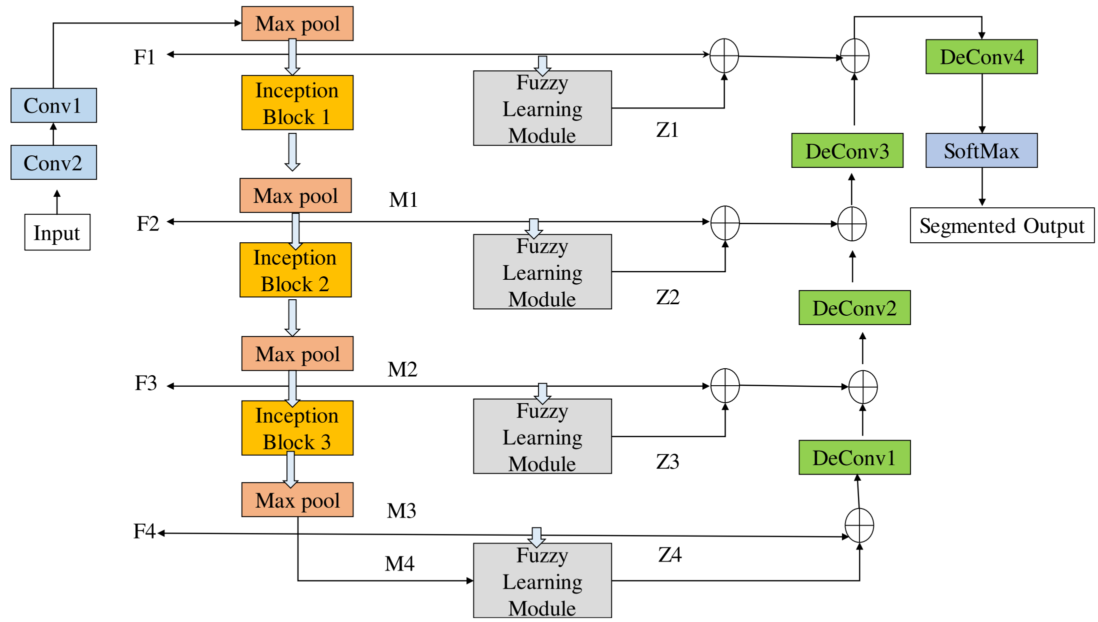

Fig. 13. LSFCNN model architecture.

The aim was to find effective solutions to uncover difficulties in accurate classification. Future research possibilities will be explored to enhance understanding of tomato leaf diseases using deep learning models.

8.1. Tackling numerous challenges in tomato leaf disease public datasets

The PlantVillage dataset faces challenges such as noise, errors, and biases. Additionally, it may have limitations in representing the various types, stages, and conditions of plant leaf diseases. These factors could affect the accuracy and generalizability. Imbalanced datasets occur when one class of data significantly outnumbers another. This leads to biased model performance, where the model tends to favor the majority class. Zhou et al. (2021) utilized the AI Challenger dataset, characterized by class imbalance, with varying image counts across classes. Notably, the class YLCV is substantially larger (4442 images) compared to other classes, for example, TS (74 images), posing challenges for model training and performance evaluation.

Given the constraints imposed by publicly accessible datasets, future research could focus on resolving class imbalances using sophisticated sampling methods. It could also aim at enhancing dataset annotation. Additionally, investigating transfer learning and domain adaptation approaches could be beneficial. Advanced sampling techniques include Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), cluster-Based Sampling, etc. For example, to generate synthetic samples for the minority class, SMOTE interpolates between instances of the minority class. This increases the representation of the minority class without exact replication. As a result, the risk of overfitting is reduced (Chawla et al. (2002)). The goal of ADASYN is to create synthetic samples by considering the density of examples of minority classes. It puts more emphasis on instances which are difficult to classify and potentially improves model's generalization capability (He et al. (2008)). Cluster-based sampling involves grouping instances into clusters and then sampling from these clusters to balance class distribution. Moreover, it guarantees that the synthetic samples produced are closer to the distribution of underlying data. Nafi and Hsu (2020) evaluated the effectiveness of GAN-based models versus traditional sampling-based approaches (undersampling, oversampling, SMOTE) for addressing class imbalance in plant disease detection. Through GAN and traditional sampling-based approaches, this study generated more samples of the minority class. According to their study, tomato leaves infected with the mosaic virus as the minority class and healthy tomato leaves as the majority class. GAN-based approach achieved higher recall, F-measure, and ROC-AUC compared to other methods. On the other hand, SMOTE-based approach indeed achieved the highest accuracy among the methods compared. Improving dataset annotation minimizes label noise, handles ambiguity of instances and mitigates biases. The goal of domain adaptation strategies is to move knowledge from a source domain with a wealth of labeled data to a target domain with limited labeled data. Some of the well-known domain adaption techniques are domain-invariant feature learning, class weighting, instance weighting, etc. Additionally, emphasizing collaboration and open data sharing facilitates knowledge exchange and accelerates progress in the field. Integration with precision agriculture technologies enhances the practicality and usability of plant disease identification models. This enables farmers to make timely and informed decisions for crop management. Fig. 15 demonstrates the strategies to handle imbalanced dataset using advanced sampling techniques, improved data annotation and domain adaption techniques.

8.2. Confronting varied challenges in image preprocessing

Numerous studies demonstrate how poorly models perform when facing complicated background clutter and varying image resolution. These factors all have an impact on feature learning (Ahmed et al. (2022); Özbılge et al. (2022)). Handling variations in image quality, resolution, and lighting conditions is one of the most challenging aspects of image preprocessing. Managing complex backgrounds is also a significant challenge. These factors greatly impact the reliability and generalization ability of classification models. Ensuring robustness across diverse environmental settings adds another layer of complexity to the preprocessing task. These challenges require careful consideration and implementation of preprocessing techniques. This enhances the quality and consistency of input data for classification models. Failure to address these issues adequately can lead to the degradation of the performance of the model in real-world scenarios.

Therefore, addressing these challenges effectively is essential for developing accurate and reliable image classification systems. Some of the novel techniques to enhance the image quality are histogram equalization, CLAHE, etc. Histogram equalization adjusts the intensity distribution of images to improve contrast and enhance details. This helps

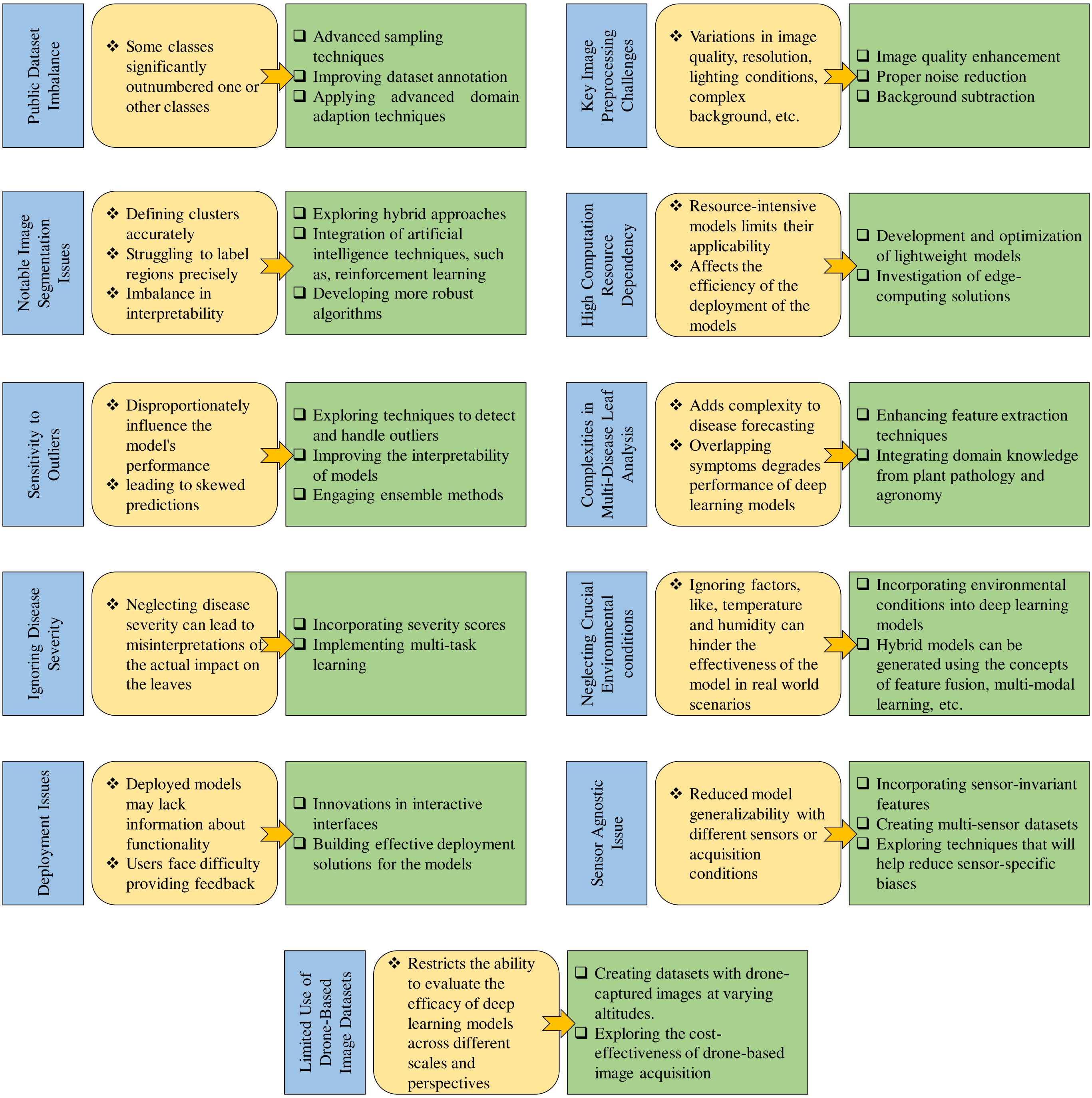

Fig. 14. Challenges and future research directions of tomato leaf disease identification.

mitigate the effects of uneven lighting conditions. On the other hand, CLAHE adaptively enhances local contrast which is particularly useful for images with varying lighting conditions. Proper noise reduction techniques, such as, gaussian smoothing, median filtering, adaptive image denoising, etc. can also be introduced. These techniques significantly reduce the noises from images. When main focus of the research is on the leaves of tomato plant, complex backgrounds can be removed by utilizing proper background subtraction techniques. Gaussian Mixture Models (GMMs), Otsu's Thresholding, U-NET, Mask R-CNN, etc. are some popular background subtraction methods. Moreover, multi-resolution techniques, for example, pyramid blending and multi-scale feature extraction can be integrated. Pyramid blending creates image pyramids at different resolutions and combines them to reduce noise and maintain features. It also enhances the overall image quality. Multi-scale feature extraction method extracts features at different scales using techniques, for example, wavelet transforms or multi-scale CNN architectures. They enable the model to capture information across varying levels of detail. Additionally, developing adaptive algorithms which are capable of robust feature extraction can significantly improve the accuracy and reliability of disease identification models. Fig. 16 demonstrates the techniques to confront numerous challenges (noise, resolution, complex background, etc) in image preprocessing. Liu et al. (2024) mentioned that images captured in natural environments often have various complex backgrounds. These backgrounds can interfere with the model's ability to learn disease and pest features. Some diseases and pests have very small and subtle characteristics on maize leaves. This makes it difficult for the model to extract these features accurately amidst complex backgrounds. The authors proposed a study on an identification method for maize leaf diseases and pests. They developed a combination of bottleneck modules with attention mechanisms. These modules help focus on disease and pest characteristics. This reduces interference from complex backgrounds. Multi-Hop Local-Feature Fusion Architecture (MLFFA) enhanced the extraction and fusion of global and local features. It effectively addressed the problem of subtle feature extraction. This approach worked well in complex backgrounds.

8.3. Navigating challenges in image segmentation techniques

Segmentation techniques are time consuming. This techniques also add an extra layer of complexity in image classification tasks. Segmentation sometimes finds it difficult to accurately define boundaries between objects (Luo et al. (2023)). Addressing the challenges in image segmentation involves various techniques each with its own set of issues. For example, U-Net requires optimization for efficient semantic segmentation. Mzoughi and Yahiaoui (2023) identified limitations in accurately identifying subtle disease symptoms in segmentation using U-Net. Deep dreams, while powerful, can present challenges in balancing interpretability and accuracy.

Instead of relying solely on one segmentation technique, researchers may combine different methods to capitalize on their individual strengths. This hybrid approach could lead to improved accuracy and adaptability in segmenting images. To enhance segmentation performance, there is a need for algorithms that can handle diverse image characteristics including variations in lighting, resolution and background. Robust algorithms would be capable of accurately segmenting images under different conditions. Researchers can integrate artificial intelligence techniques such as machine learning and reinforcement learning. This integration pushes the boundaries of image segmentation capabilities. These techniques could enable algorithms to learn and adapt to different types of images segmentation tasks. Lastly, continuous research on novel algorithms and the incorporation of real-time feedback mechanisms could contribute to the ongoing evolution of image segmentation methodologies. Hasan et al. (2022) developed a novel lab color space-based segmentation method. This method doesn't require image preprocessing and can effectively handle uneven lighting and exposure. The authors also proposed an innovative and modified feature extraction method. This method focuses on extracting Discrete Wavelet Transformation (DWT) coefficients. These coefficients are less susceptible to noise. Subsequently, they combined DWT features and lab color histogram features horizontally to create a robust feature vector. Their proposed model achieved an impressive accuracy of 98.63 % in detecting apple leaf diseases using the Random Forest classifier. Fig. 17 represents the development of a robust model to navigate challenges in segmentation following some protocols, such as cleaning the dataset, model monitoring and model verification.

8.4. Probable solutions of high computational resource dependency

The computational demands of classification and detection models pose significant challenges. A study by Jing et al. (2023) revealed that these challenges create obstacles to accessibility, scalability, and widespread deployment. These resource-intensive models limit their applicability across diverse sectors. They also impede the rapid pace of innovation. The substantial computational requirements act as a barrier which is hindering the broader adoption and integration of these models in the real-world scenarios. This limitation not only affects the efficiency of deployment but also restricts the potential impact of these models on various industries. Addressing these computational challenges is crucial to unlock the full potential of classification and detection models for broader and more practical use.

The development and optimization of lightweight models may become the primary focus in the future. Lightweight models are computationally efficient and have a small memory footprint. This makes them suitable for deployment on resource-constrained devices such as mobile phones, IoT devices, or edge devices. Traditional deep learning models can be computationally expensive and require large amounts of memory and processing power. This may not be feasible for deployment in certain environments or on devices with limited resources. To address this challenge, researchers can explore techniques to optimize models for efficiency, such as model compression, pruning, quantization, etc. These techniques aim to reduce the size of the model and the computational resources required for inference. At the same time, they preserve the model's accuracy and performance. In addition, edge-computing solutions and the development of hardware technologies should also be investigated. Wang et al. (2021) proposed a method combining

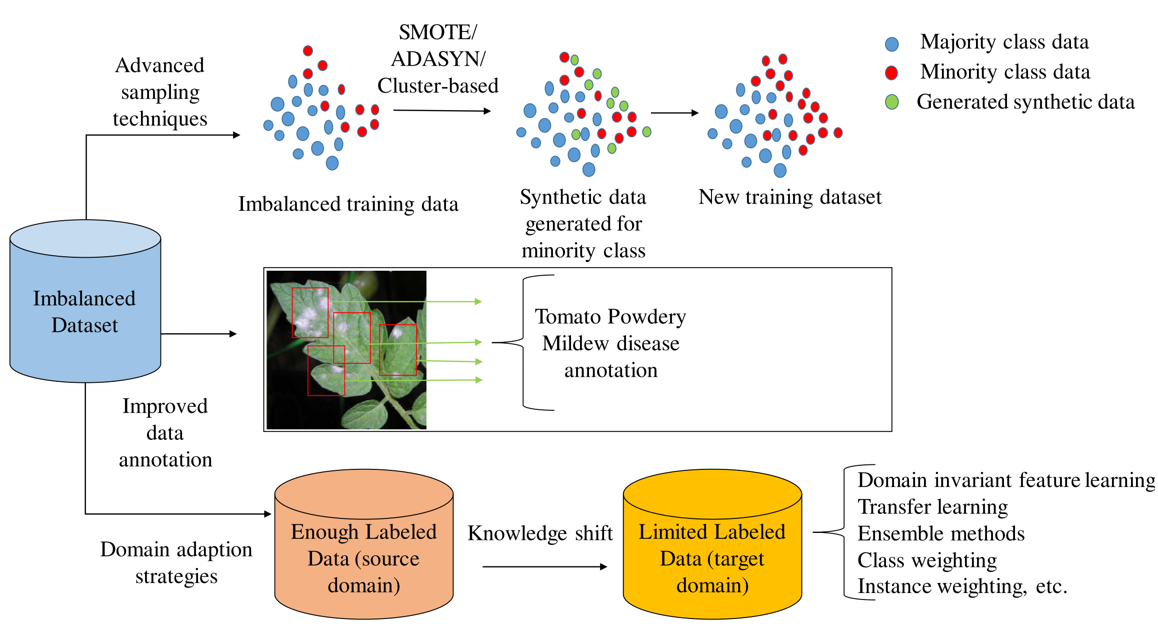

Fig. 15. Strategies to handle imbalanced dataset.

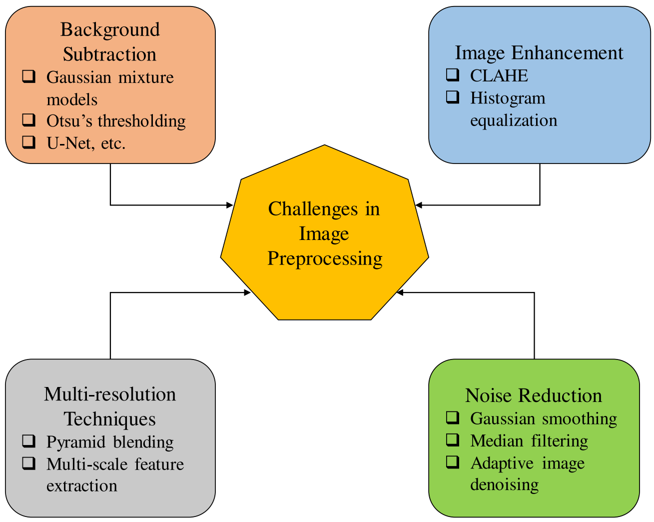

Fig. 16. Methods to address a variety of image preprocessing challenges, such as noise, resolution and complex background.

pruning, knowledge distillation, and quantization to compress deep neural networks. This approach makes the networks more efficient for deployment in resource-constrained areas. The authors applied the compressed models to the Plant Village dataset. Their results demonstrated the models' effectiveness in recognizing plant diseases. They also highlighted the benefits of using lightweight fully connected layers and iterative pruning to reduce computational burden and improve model efficiency. The compressed models demonstrated high accuracy (97.09 %) while significantly reducing the model size to 0.04 MB. Edge computing involves processing data locally on edge devices rather than sending it to a centralized server or cloud. This approach can reduce latency and bandwidth usage. Optimizing models for edge deployment requires considering the constraints of edge devices. These constraints include limited processing power, memory, and energy consumption. Model quantization is used to optimize a deep learning model for deployment on hardware with constrained computational resources. It involves decreasing the precision of numerical variables in the model. Whereas reducing the size of a deep learning model by eliminating unnecessary parameters, connections, or weights is known as model pruning. This technique helps make the model more efficient.

8.5. Strategies to handle outliers

Data are formed by various generational processes or observations gathered from one or more entities during the creation of AI-based systems. An outlier point is created by an individual or group of entities exhibiting anomalous behavior. Therefore, in order to assess the health of a system, it is crucial to comprehend the behavior of outliers. This understanding helps anticipate future system breakdowns (Sikder and Batarseh (2023)). Sensitivity to outliers refers to the susceptibility of a statistical model or algorithm to extreme or unusual data points. These data points are commonly referred to as outliers. Outliers are data points that deviate significantly from the majority of the data. They can distort the statistical properties of a dataset. In the context of modeling, outliers have the potential to disproportionately influence the model's performance. It also leads the model to skewed predictions and affecting the overall accuracy and robustness of the model.

Subsequent investigations could concentrate on developing algorithms that are less affected by individual extreme data points. These robust algorithms would be better equipped to handle outliers without significantly affecting their performance or predictions. Exploring techniques especially to detect and handle outliers could be another future research. Models can be strengthened against the impact of outliers by correctly recognizing and handling them. Future research could also aim to improve the interpretability of models to better understand and mitigate the impact of outliers. By gaining insights into how outliers affect model predictions, researchers can develop strategies to mitigate their influence more effectively. Ensemble methods involve combining multiple models to improve overall performance. Future research may explore ensemble techniques as a means of building more capable models in the presence of outliers. By aggregating predictions from multiple models, ensemble methods can help reduce the impact of outliers on individual model predictions. Advancements in machine learning techniques, such as, deep learning or reinforcement learning can also contribute to the development of more robust models in the presence of outliers. These advanced techniques may offer novel approaches for outlier detection, handling, and model training. This leads to improved performance in outlier-prone datasets. Fig. 18 demonstrates some general and advanced techniques to handle the outliers. General techniques include deletion, imputation, transformation, and robust regression. On the other hand, advanced techniques include ensemble

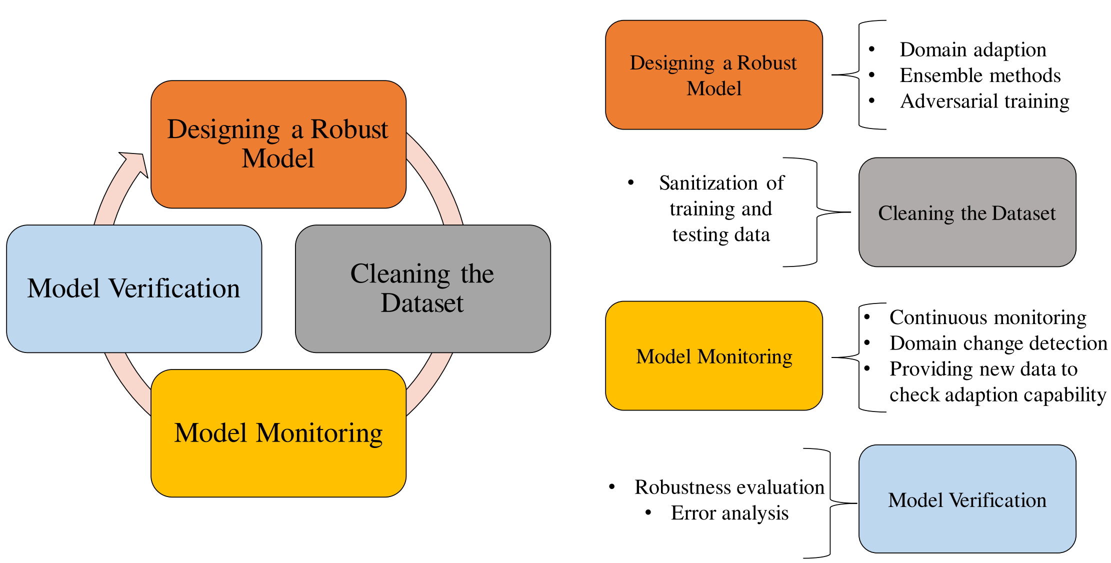

Fig. 17. Development of a robust model following some protocols to navigate challenges in segmentation process.

methods, reinforcement learning, improving interpretability, and shifting domain knowledge. Usha Ruby et al. (2024) introduced a Collaborative GAN for image imputation. This approach helped in accurately estimating the missing data. This process ensured that the dataset remained complete and improved the quality of the images used for training the model. By doing so, the imputation step enhanced the feature extraction process. This led to more accurate classification of wheat leaf diseases. Deletion involves removing outliers from the dataset. While imputation means replacing the outlier with a more representative value of the dataset. On the other hand, transformation means transforming the dataset so that it reduces the overall impact of outliers. Robust regression is a model which is designed to handle the outliers of a dataset.

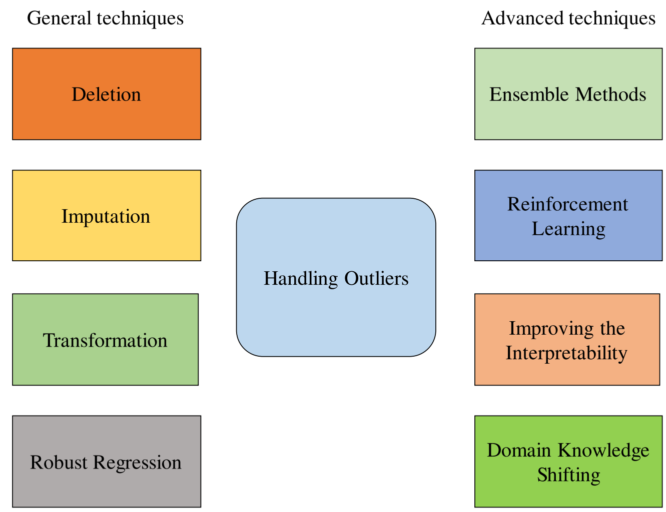

Fig. 18. General and advanced techniques to handle outliers.

8.6. Complexities in multi-disease leaf analysis and possible solutions

Analyzing entire leaf bunches adds complexity to disease forecasting. Leaf bunches may contain multiple leaves each potentially affected by different diseases or exhibiting varying stages of disease progression. This complexity increases the difficulty of accurately identifying and distinguishing between different diseases (Sahu et al. (2023)). In addition to the complexity introduced by analyzing leaf bunches, the presence of multiple diseases on a single leaf adds further complications. This makes disease forecasting even more challenging. Different diseases may manifest with overlapping symptoms or affect different parts of the leaf. This makes it more challenging for traditional methods. Even some deep learning models face challenges to accurately identify and differentiate between the diseases.

Enhancing the extraction of features from images of tomato leaves could lead to more discriminative and informative representations. Advanced feature extraction techniques, such as deep learning-based approaches or feature fusion methods, may be explored. These techniques can help capture subtle patterns associated with different diseases accurately. Fig. 19 demonstrates the architecture of multi-level feature fusion method to handle the complexities in multi-disease leaf analysis. Multi-level feature fusion can integrate features extracted at different levels of abstraction, such as texture, shape and color from leaf images. It will enable a comprehensive representation of disease-related patterns. This will enhance classification accuracy across multiple diseases. Understanding the complex relationships between various diseases is crucial since numerous diseases might coexist on a single leaf. Researchers may utilize innovative techniques, for example, network analysis or graph-based methodologies. These approaches help clarify the complex connections and interdependencies among various illnesses. This enables more precise disease detection and prediction. Integrating domain knowledge from plant pathology and agronomy can enhance the interpretability and effectiveness of disease identification models. Researchers can incorporate expert knowledge about disease symptoms, host-pathogen interactions, and environmental factors. This helps them develop more informed models for disease forecasting. These models are also more contextually relevant for tomato plants.

8.7. Ignoring disease severity and impact on tomato leaf disease diagnosis

Disease severity refers to the intensity of the disease's impact on the plant. It is a crucial factor that can profoundly influence the observable symptoms and overall health of the plant. However, many current models predominantly rely on observable symptoms without considering the severity of the disease. This can compromise the accuracy of diagnoses (Mzoughi and Yahiaoui (2023); Cheng et al. (2022)). Neglecting disease severity can lead to misinterpretations of the actual impact on the leaves. For example, two leaves may exhibit similar symptoms, such as discoloration or lesions but one leaf may be more severely affected than the other. Without considering disease severity, a model may inaccurately classify both leaves as having the same disease. However, the severity of the disease may differ significantly between them.

Future research can be extended to develop methods to quantify disease severity and integrating severity scores as additional features or labels in the training data. This would enable models to learn the relationship between observable symptoms and disease severity. It will improve their ability to make accurate diagnoses. Multi-task learning techniques can enable models to use shared representations between tasks. This will improve performance on both disease identification and severity estimation. Fig. 20 demonstrates disease severity prediction along with certain tomato leaf disease by a method diagram which employs multi-task deep learning model. By addressing the critical consideration of disease severity, future deep learning models for tomato leaf disease identification can improve the accuracy and reliability of diagnoses. It may lead to more effective plant disease management strategies and ultimately will contribute to enhanced crop yield and food security. Pal and Kumar (2023) proposed a framework for detecting plant diseases and classifying their severity using image processing techniques. The framework integrated Kohonen-based deep learning with a pre-trained INC-VGGN model for accurate disease detection and classification. But the authors faced high cost complexity, misclassifications, and overfitting problems in real-time applications.

8.8. Neglecting crucial environmental factors and impact on tomato leaf disease diagnosis

The current limitation of many deep learning models is neglecting crucial environmental conditions, such as temperature and humidity. These environmental factors play a vital role in influencing the growth and spread of pathogens responsible for leaf diseases in tomatoes. Ignoring these factors can lead to inaccurate predictions and hinder the effectiveness of disease prediction models in real-world scenarios. Temperature and humidity are the key environmental variables that directly impact the development and severity of plant diseases. For example, many fungal pathogens thrive in warm and humid conditions while others may prefer cooler temperatures. Likewise, variations in humidity levels can affect the rate of pathogen reproduction and spore dispersal which ultimately influence disease spread.

Incorporating environmental conditions such as temperature and humidity into deep learning models for disease prediction is essential. This approach enhances the models' accuracy and reliability. By considering these factors, models can better capture the complex interactions between environmental conditions and

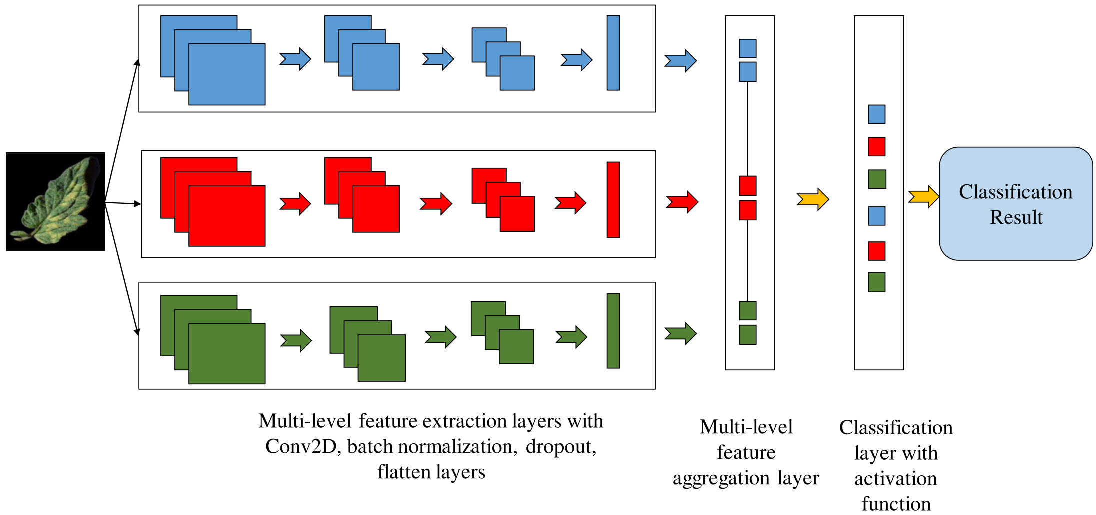

Fig. 19. Multi-level feature fusion architecture to handle the complexities in multi-disease leaf analysis.

disease development. This leads to more precise predictions and actionable insights for disease management. Integrating environmental modules into existing deep learning architectures is a promising approach. It explicitly models the influence of environmental conditions on disease prediction. This involves designing hybrid models that combine deep learning approaches with domain-specific knowledge about environmental factors and disease dynamics. These hybrid models can be generated using the concepts of feature fusion, multi-modal learning, attention mechanisms, etc. For example, Zhang and Chen (2023) developed a Smart Farm Monitoring System (SFMS). Machine learning and deep learning were used to enhance the detection and prevention of leaf curl disease in tomato crops. These algorithms analyze the collected data (images of tomato leaves, temperature, humidity, soil moisture) to identify patterns and anomalies associated with leaf curl disease. These algorithms are trained to recognize the visual symptoms of leaf curl disease from the images. This enables early and accurate detection. By analyzing real-time data, the system can predict potential outbreaks of the disease. It can also suggest preventive measures to farmers, improving crop health and yield. This SFMS can be further refined to detect and manage multiple tomato leaf diseases. Incorporating novel deep learning architectures could significantly enhance the model's performance. This would improve the system's effectiveness in disease detection and management. Fig. 21 demonstrates a smart system to incorporate environmental data enabling the model to predict a disease that can occur in tomato leaves in the farm. This system includes humidity sensor, temperature sensor and drone to collect environmental data. Deep learning model will be trained on previously collected data responsible for a certain disease to occur. The sensors and the camera will continuously send data to the trained model. The data will be processed through a microcontroller. This will trigger the model to detect certain tomato leaf diseases.

8.9. Enhancing deployment strategies and implementing interactive interfaces

Many deep learning models that has been deployed for tomato leaf disease identification on web platforms offer limited interaction options for users. Users may have difficulty providing feedback, asking questions or accessing additional information relevant to their specific needs or concerns. The deployed models may lack

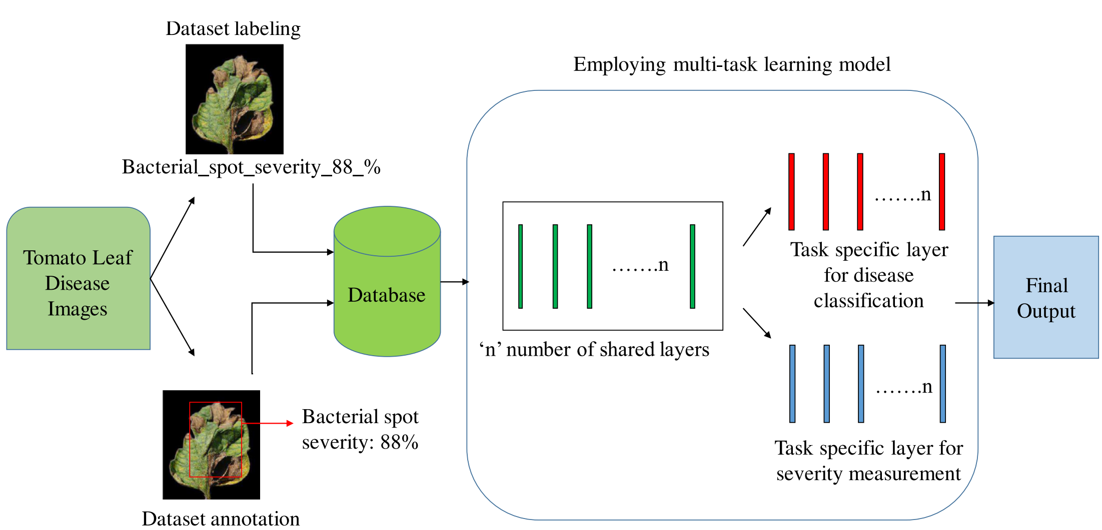

Fig. 20. Disease severity prediction along with certain tomato leaf disease: a method diagram employing multi-task deep learning model.

comprehensive information about their functionality and performance metrics. They also may not provide insight into their limitations and the underlying decision-making process (Paul et al. (2023)). Without access to this information, users may struggle to understand how the model works and how confident they can be in its predictions.

To address this, future research should prioritize comprehensive studies of real-time deployment strategies with a strong emphasis on enhancing user experience. Incorporating feedback mechanisms for continuous improvement of the model can be another active area of research. Innovations in user-centric design and interactive interfaces hold great potential. They can significantly enhance the usability and effectiveness of deployed models in practical scenarios. By fostering closer alignment with end-user needs and preferences, these advancements can pave the way for more impactful applications of deep learning models. They can also facilitate seamless interaction and engagement in real-world settings. Fig. 22 represents the proper interaction between the model and the end-user after deployment of the model.

8.10. Sensor agnostic issue

Deep learning models for image processing are often trained on data collected from specific sensors or imaging devices. These devices, such as particular cameras or imaging setups, are used in controlled environments. This dependency introduces a sensor-agnostic issue. Models perform well on data from the original sensor but may fail to generalize. When applied to images from different sensors, the models struggle due to varying characteristics, such as resolution, color spectrum, or noise levels. This issue is particularly relevant in agricultural applications, where images may be captured under diverse field conditions with different devices. The sensor agnostic problem can limit the practical deployment of deep learning models in real-world agricultural settings. For example, a model trained on high-resolution images from a laboratory setting may not perform adequately. When applied to lower-resolution images captured by a farmer's smartphone, its performance may decrease. This discrepancy reduces the model's robustness and restricts its utility across different environments and users. Deep learning models are highly sensitive to variations in environmental conditions, such as lighting, angle of capture, and image resolution. These factors can vary widely depending on the sensor used. As a result, they lead to inconsistent performance. For example, a model trained on images captured in optimal lighting conditions may struggle with images taken in low light or with shadows. These conditions are common in field settings.

Researchers must prioritize enhancing the robustness of deep learning models to sensor variations. This can be achieved by training on diverse datasets. Incorporating sensor-invariant features helps minimize dependence on specific imaging devices. The creation of standardized, multi-sensor datasets is crucial for training models that perform consistently across varying imaging conditions. Such datasets should include a broad spectrum of sensor types and environmental conditions, accurately reflecting real-world variability. By doing so, models can better generalize and maintain accuracy across different settings. This makes them more reliable and applicable in practical scenarios. Additionally, exploring advanced techniques to reduce sensor-specific biases during training will further enhance model robustness. Ultimately, these efforts will lead to the development of more versatile and reliable deep learning models. These models will be capable of maintaining high performance despite variations in sensor technology or environmental factors.

8.11. Limited use of drone-based image datasets

Drone-based image acquisition offers potential advantages. However, there is a noticeable shortage of datasets that use this method, particularly for tomato leaf disease detection. Most existing studies rely on close-up images taken at the leaf level. These images are primarily captured using mobile phones or handheld cameras. The lack of research utilizing drone-captured images at varying distances creates a significant gap. It limits our understanding of how these images can be effectively used in deep learning models for agricultural applications. The scarcity of drone-based datasets restricts the ability to evaluate the efficacy of deep learning models across different scales and perspectives. This limitation highlights a critical area requiring further research. Additional studies are needed to develop more robust and versatile image processing techniques.

Researchers should prioritize the creation of comprehensive datasets that include images captured by drones at different altitudes and distances. These datasets should encompass a variety of environmental conditions to enhance the generalizability of deep learning models. Investigating the impact of varying distances and perspectives

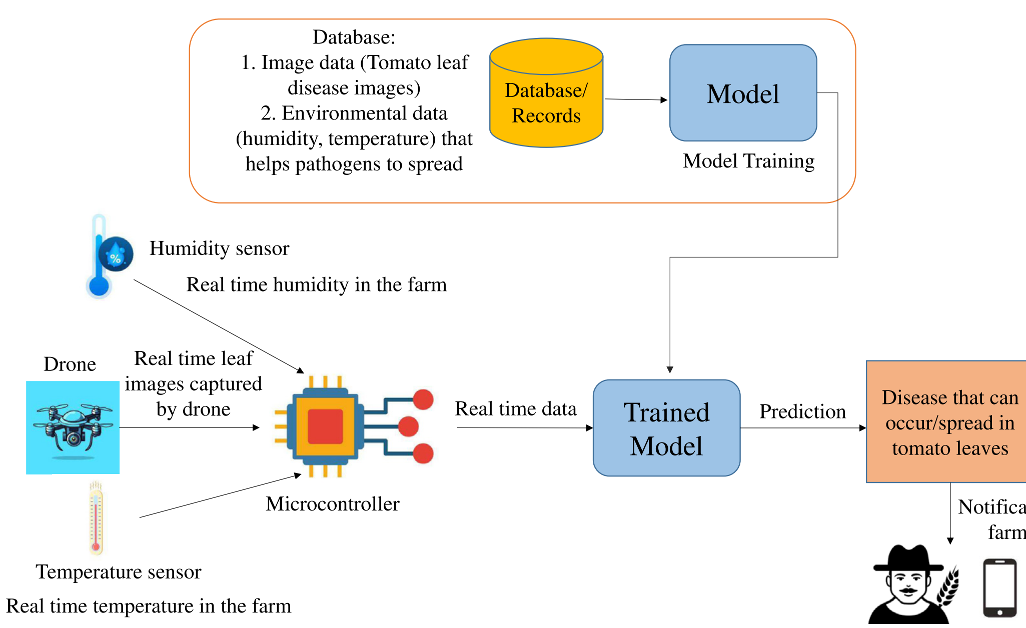

Fig. 21. A smart system to incorporate environmental data enabling the model to predict a disease that can occur in tomato leaves in the farm.

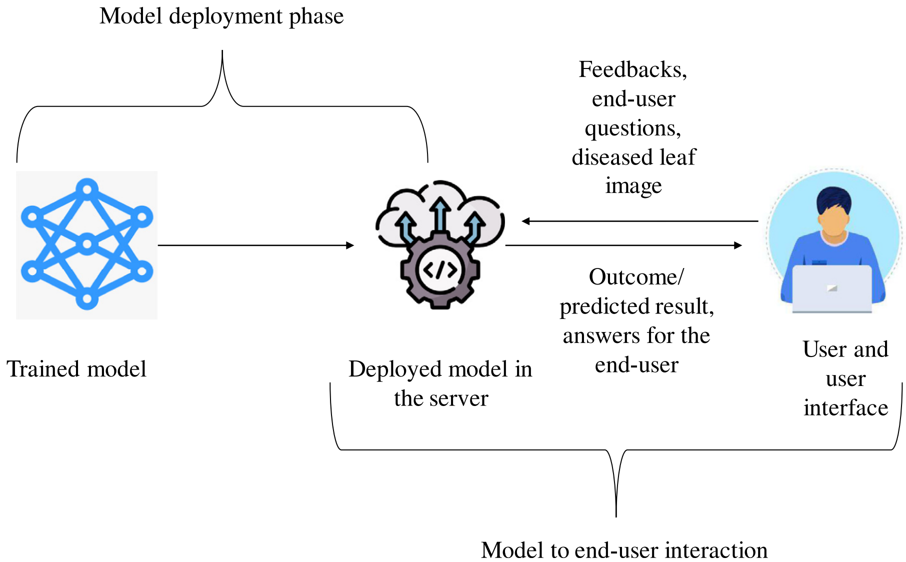

Fig. 22. Post deployment phase: interaction between the model and the end-user.

on model accuracy is essential. Future studies should conduct comparative analyses of models trained on close-up and drone-captured images. They should examine how the distance to the target affects detection and classification performance. Research should also explore the cost-effectiveness of drone-based image acquisition compared to traditional methods. Assessing the trade-offs between image quality, model performance, and economic factors is essential. This evaluation will help determine the practicality of deploying drones in real-world agricultural settings. Consider combining mobile phone and drone images within the same dataset to create hybrid models. Such models could combine the strengths of both methods. This approach would improve accuracy and robustness across different scenarios and conditions.

9. Conclusions

In the field of identifying tomato leaf diseases using deep learning, a comparative study provides a detailed look at the situation. This research explored various models, datasets and evaluation metrics to provide a complete picture of the use of deep learning for disease identification. Its importance lies in showing how different factors affect deep learning models when dealing with tomato leaf diseases. This study highlights the need for flexible solutions to understand the specific features of each disease. In order to increase the models' accuracy and resilience, it highlights the difficulties associated with deep learning and promotes continuous development. This study is a valuable guide for future research, urging innovative approaches and a thorough understanding of plant diseases. Ultimately, this exploration contributes to discussions on agricultural technology. It guides the development of more effective deep learning systems for identifying tomato leaf diseases. The insights gained from this study can lead to advancements in plant disease identification in agriculture and technology.

CRediT authorship contribution statement

Aritra Das: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Fahad Pathan:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Jamin Rahman Jim:** Visualization, Validation, Resources, Methodology. **Md Mohsin Kabir:** Writing – review & editing, Resources, Investigation, Formal analysis. **M.F. Mridha:** Writing – review & editing, Validation, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aiia.2025.02.006.

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